

claude-windows / da34752f-c664-4262-9602-f5d4cfcc6db4

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_cffc9fdf-8f2\agent-a8b43cc374f30224b.jsonl`  
- SHA-256: `caeb8fab43695d30fbdd255dc682ce338435a395cf33719ca97b06ac710e022`  
- Source modified: `2026-06-12T09:07:11+00:00`  
- Imported at: `2026-07-05T16:48:23+00:00`  
- project: `wf_cffc9fdf-8f2`  
- session_id: `da34752f-c664-4262-9602-f5d4cfcc6db4`
```

Transcript

1. user (2026-06-12T09:03:26.884Z)

The three sibling repos (all are accessible working directories):

- C:\Users\avidu\Projects\badminton-highlight-indexer (main indexer: FastAPI + ffmpeg pipeline)
- C:\Users\avidu\Projects\sports-data-collector (data collection; believed to hold the Roora vendor ingestion flow)
- C:\Users\avidu\Projects\rally-annotator (VLC-based annotation tool)

TASK: Find every place where the Roora annotation vendor's timing delay/offset is documented. The user believes "their annotated videos are almost always 0.5+ seconds delayed" and that this is documented somewhere.

Search broadly across ALL THREE repos (docs/, scratch/, memory notes, code comments, eval scripts, README files):

- Keywords: Roora, vendor, delay, offset, lag, shift, late, 0.5, 500ms, systematic, bias, tolerance, IoU, boundary
- Look in golden-set / vendoring docs especially (the badminton-highlight-indexer has docs about golden-set vendoring with Roora; GS-1/GS-2/GS-4 landed, GS-3 pending)
- Also check eval/comparison scripts that compare vendor annotations to detector output or to a gold standard ? there may be measured offset numbers.

Report: WHERE the delay is documented (exact files+lines), the exact measured numbers (mean/median offset, direction ? are rally starts late, ends late, or both?), any documented hypothesis for the CAUSE (e.g. their tool seeks only to keyframes, plays at low fps, human reaction time, VFR video), and any documented mitigation discussed before (e.g. post-hoc offset correction, tolerance windows in eval).

2. assistant (2026-06-12T09:03:29.052Z)

I'll search across all three repos systematically for documentation of the Roora vendor timing delay/offset. Let me start with broad searches for key terms and then dig into specific areas.

3. user (2026-06-12T09:03:29.663Z)

```
.pytest_cache\README.md  
docs\archives\README.md  
docs\archives\checkpoints\2026-06-eval-baseline\README.md  
docs\archives\checkpoints\2026-06-logging-reporting\README.md  
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md  
docs\archives\past_projects\README.md  
docs\archives\checkpoints\2026-06-strategy-foundation\README.md  
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md  
docs\archives\quality-iterations\2026-06-initial\README.md  
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md  
docs\archives\quality-iterations\README.md  
docs\README.md
```

training\README.md
README.md

4. user (2026-06-12T09:03:29.907Z)

docs\EVALUATION.md
docs\GOLDEN_SET_IMPLEMENTATION.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\archives\README.md
docs\PMF_MARKET_RESEARCH.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\past_projects\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\VIDEO_RECORDING_GUIDELINES.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
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docs\archives\quality-iterations\README.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
.git\logs\refs\remotes\origin\docs\next-steps-multisport
.git\refs\remotes\origin\docs\next-steps-multisport
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\COMMERCIALIZATION.md
docs\GOLDEN_SET_VENDORING.md
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs\PLATFORM_ARCHITECTURE.md
docs\README.md
docs\ROADMAP.md
docs\TRACKNET_WSL_SETUP.md
docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md
docs\HOSTING_PLAN.md
docs\UI_PROPOSAL.md
docs\archives\decisions\DECISIONS.md
docs\mockups\ui_alpha_mockup.html
docs\archives\TEST_RUN_SUMMARY.md
docs\CODE_MAP.md
docs\DEFERRED_HARDENING_PLAN.md
.git\logs\refs\remotes\origin\docs\pr2-merge-status-sync
.git\refs\remotes\origin\docs\pr2-merge-status-sync
docs\BACKLOG.md
docs\NEXT_STEPS.md
docs\UI_ALPHA_EXECUTION_PLAN.md

5. user (2026-06-12T09:03:29.944Z)

.pytest_cache\README.md
docs\EVALUATION.md
docs\GOLDEN_SET_IMPLEMENTATION.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\HOW_RALLY_DETECTION_WORKS.md
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docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
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docs\archives\quality-iterations\README.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
scratch\HIGH_RISK_REMEDIATION_PLAN.md
CLAUDE.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
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docs\PLATFORM_ARCHITECTURE.md
docs\README.md
docs\ROADMAP.md
docs\TRACKNET_WSL_SETUP.md
training\README.md
output\kushagra_1280.report.md
output\kushagra_1280.summary.md
docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md
scratch\PR1_HANDOFF.md
scratch\PR1_BODY.md
scratch\PR88_FIX.md
docs\HOSTING_PLAN.md
docs\UI_PROPOSAL.md
docs\archives\decisions\DECISIONS.md
docs\archives\TEST_RUN_SUMMARY.md
README.md
output\mahadevpura_smoke_180s.report.md
output\mahadevpura_smoke_180s.summary.md
scratch\PR2_HANDOFF.md
docs\CODE_MAP.md
docs\DEFERRED_HARDENING_PLAN.md
docs\BACKLOG.md
docs\NEXT_STEPS.md
docs\UI_ALPHA_EXECUTION_PLAN.md

6. user (2026-06-12T09:03:30.340Z)

Found 11 files

docs\NEXT_STEPS.md
docs\BACKLOG.md
docs\archives\decisions\DECISIONS.md
docs\COMMERCIALIZATION.md
docs\GOLDEN_SET_VENDORING.md
docs\ROADMAP.md
scratch\HIGH_RISK_REMEDIATION_PLAN.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\GOLDEN_SET_IMPLEMENTATION.md

7. assistant (2026-06-12T09:03:32.625Z)

Excellent! I found the key file: `docs\GOLDEN\_SET\_VENDORING.md`. Let me read that and the other Roora-mentioning documents:

8. user (2026-06-12T09:03:33.029Z)

```

1      # Golden-Set Generation & Vendor Process (v1)
2
3      **Status:** Design v1 . **Date:** 2026-05-30 . **Vendor shortlist added 2026-05-30
($6)**
4      **Owner decisions captured:** licensed/owned video only . rally `[start,end)` depth .
tiered turnaround (fast automated + slow human) . vendor bake-off . **Bangalore/India-local
vendor preference**
5
6      This document defines how we produce **golden-set ground truth** (hand-verified rally
boundaries) for the badminton highlights product, how we **select annotation vendors**, and how
that same machinery powers **on-demand / surprise regression testing** of the production tool.
7
8      > **Companion docs:** the scoring mechanics live in [EVALUATION.md](EVALUATION.md); the
annotated-corpus rationale is [ADR-002](archives/decisions/DECISIONS.md); the decision record
for *this* process is **ADR-006**. Video-capture quality requirements (a dependency of
trustworthy golden sets) are tracked separately in
[VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md).
9
10     > **Update (2026-06-02):** $6 shortlist done; the founder selected **Roora** (Bengaluru)
for the **initial paid pilot** ? a **single-vendor pilot first**, not the full bake-off below.
Roora's site/capability/location are verified; **quality is TBD via the pilot**. **Step 0 (the
annotation guideline) is authored** ? collector repo `docs/annotation/` (`ROORA_BRIEF.md` rally
CSV schema + `ANNOTATION_GUIDE.md` illustrated guide + intake-manifest & NDA templates). Pilot
scope is **3 sports** (badminton, pickleball, table tennis); the collector's `import-local`
ingestion for all three is built. Still pending: the internal **gold key** and the vendor-
dependent **GS-3 `HumanVendorProvider`** (GS-1/GS-2/GS-4 already landed ? see
[GOLDEN_SET_IMPLEMENTATION.md](GOLDEN_SET_IMPLEMENTATION.md)).
11
12     ---
13
14     ## 1. The keystone insight: one scorer, three jobs
15
16     A vendor's golden-set output is a list of rally `[start,end)` intervals ? **structurally
identical to a prediction** from our pipeline. So the *exact same* deterministic scorer in
`backend/eval/metrics.py` (`interval_iou` ? `match_intervals` ? `score`) grades all three of:
17
18     1. **Product vs. ground truth** ? the precision/recall/F1/IoU we already report
(`EVALUATION.md`).
19     2. **Vendor vs. our internal answer key** ? the quality gate during vendor selection.
20     3. **Annotator vs. annotator** ? inter-annotator agreement (IAA), our proxy for "is this
even a well-defined task?"
21
22     This symmetry is what makes the whole system small, auditable, and extensible. **No new
scoring code is needed** ? only a new *source* of intervals (the vendor).
23
24     ---
25
26     ## 2. Why this also solves the privacy nuance
27
28     The monetization goal (hosted API) carries a "someone views the user's video" concern.
This process sidesteps it **by construction**:
29
30     - **Golden sets are built only from footage we license or own** ? starting with the
founder's **constantly-growing personal collection** of match videos, plus ShuttleSet (ADR-003)
and any purpose-recorded clips. **No end-user uploads ever flow to a vendor.**
31     - Regression tests therefore run against a **representative held-out corpus**, not
literal user content.
32     - All video egress to any third party passes through a **single provider boundary**
($4), so the security controls (DPA, no-reuse, delete-on-delivery, access logging) live in
exactly one place.
33
34     > **Honest tradeoff:** a licensed/owned corpus may not perfectly mirror real users'
cameras, courts, and lighting. A green regression on our corpus is therefore *strong evidence*,
not a *guarantee*, for production traffic. Revisit if field behavior diverges from corpus
behavior ? and see VIDEO_RECORDING_GUIDELINES.md, which aims to shrink that gap by steering

```

```

users toward capture conditions our corpus represents.
35
36     ---
37
38     ## 3. Selecting vendors ? the bake-off
39
40     ### Step 0 ? Write the annotation guideline (highest-leverage artifact)
41     Before contacting anyone, write a precise, example-rich definition of what a rally
is:
42     - Start trigger: e.g. first serve motion vs. shuttle leaving the racket ? pick one,
show clips.
43     - End trigger: shuttle hits floor/net/out, or umpire call.
44     - Edge cases: lets/replays, service faults, between-point walking, coaching pauses,
mid-rally net-cord, broadcast replays and graphics.
45
46     Rally boundaries are genuinely subjective; ambiguity here is the #1 cause of bad
golden sets. This guideline belongs conceptually with the per-sport SportProfile, so it
generalizes to tennis/cricket later.
47
48     > Done (2026-06-02): authored as the Roora brief + illustrated guide in the
collector repo docs/annotation/. Defines the rally CSV schema and the three golden rules ?
ignore all dead time; include every rally (incl. service faults & lets); a shot = one
racket/paddle/bat contact ? plus an ending_reason taxonomy (winner / forced_error /
unforced_error / service_fault / let / other, shared across net-separated sports) and per-sport
detail incl. singles & doubles for badminton & pickleball.
49
50     ### Step 1 ? Build a small internal gold key
51     One trusted expert (initially: the founder) hand-annotates 5?10 licensed clips to
near-perfect. This is the answer key vendors are graded against ? kept private during the bake-
off.
52
53     ### Step 2 ? Run the bake-off (parallel pilot)
54     Send the same pilot clips to 2?3 shortlisted vendors (see §6, favoring
Bangalore/India), withholding the key. Grade every submission with our own harness:
55
56     | Dimension | How we measure it | Tool |
57     |---|---|---|
58     | Accuracy | vendor CSV vs. gold key ? P/R/F1, mIoU, boundary error | eval.score()
|
59     | Consistency | 2 annotators per vendor ? pairwise IoU (IAA) | interval_iou |
60     | Cost | quoted ? or $ per video-minute | vendor quote |
61     | Turnaround | actual vs. promised, per tier | observed |
62     | Ops friction | file-drop vs. API; compliance with our CSV format |
annotations.load_annotations() |
63     | Security | DPA signed; DPDP/ISO posture; delete-on-delivery | contract review |
64
65     ### Step 3 ? Decide on measured quality-per-dollar
66     The bake-off numbers are the decision ? not sales decks. Low IAA at a vendor means
either a weak vendor or a vague guideline (Step 0); investigate before blaming the vendor.
67
68     ### Step 4 ? Onboard the winner(s)
69     Sign DPA + the security controls in §4; codify the guideline as their spec; agree SLAs
for both tiers (§5).
70
71     ---
72
73     ## 4. The extensible architecture ? an AnnotationProvider registry
74
75     This codebase already uses pluggable registries (validators, segmenters, AI providers,
sports). The idiomatic move is the same shape for annotation sourcing:
76
77     ```python
78     class AnnotationProvider(ABC):
79         def submit(self, video_ref, guideline, sport) -> str:           # returns job_id
80         ...

```

```

81         def poll(self, job_id) -> str:                                     #
queued|running|done|failed
82         ...
83         def fetch(self, job_id) -> List[Interval]:                         # canonical
[(start,end), ...]
84         ...
85         ...
86
87     `fetch()` returns the same `List[Interval]` that `annotations.load_annotations()`
already produces, so everything downstream is unchanged.
88
89     Three implementations cover today plus both tiers:
90
91     | Provider | Role | Tier |
92     |---|---|---|
93     | `FileProvider` | Reads a pre-made CSV (formalizes today's manual / ShuttleSet flow) |
? |
94     | `HumanVendorProvider` | The bake-off winner; authoritative ground truth, corpus
growth, deep audits | Slow (days) |
95     | `AutomatedProvider` | An independent strong model labels rallies programmatically |
Fast (hours) |
96
97     The tiers are just two providers behind one interface, selectable per call.
98
99     ---
100
101     ## 5. The regression automation (the explicit ask)
102
103     > "given a video, get a golden set created and verify its precision and recall against
the production tool"
104
105     This becomes a thin wrapper over machinery that already exists:
106
107     ```python
108     def regress(video, tier="fast", tolerance=0.05):
109         gt          = annotation_registry.get(tier).fetch(                # NEW: the provider
110             annotation_registry.get(tier).submit(video, GUIDELINE, sport))
111         predictions = process(video)                                     # existing production
pipeline ? DB
112         s          = eval.score(predictions, gt)                        # existing metrics.py,
unchanged
113         regressed  = (s.recall < baseline[video].recall - tolerance or
114             s.precision < baseline[video].precision - tolerance)
115         if regressed: alert(video, s, baseline[video])
116         return s
117     ```
118
119     Only one new component (the provider). `run_eval.py`, `harness.py`, and `metrics.py`
are reused verbatim. Snapshot a `baseline.json` per corpus video so "regression" is a measured
delta, not a vibe.
120
121     Triggers (all feed the same `regress()`): CI on pipeline changes · on-demand (the
"surprise" case) · scheduled cron · pre-release gate.
122
123     ### ?? The oracle problem (a real correctness trap)
124     The fast-tier `AutomatedProvider` must be a different model lineage than production**.
If production becomes the offline ActionFormer segmenter (ADR-004) and the oracle is also
ActionFormer, they share blind spots ? the test shows green while both are wrong. Pairings that
stay honest:
125
126     - production = offline learned segmenter ? fast-tier oracle = paid Gemini full-video
(different lineage).
127     - production = Gemini hybrid ? fast-tier oracle = a different VLM or the slow human
tier.
128

```

129 Treat the fast tier as a **smoke alarm**, not ground truth: any fast-tier "regression"
escalates to the slow human tier for confirmation before it blocks a release.

130

131 ---

132

133 ## 6. Vendor shortlist (Bangalore/India-favoring)

134

135 > **Researched 2026-05-30 via a lean manual pass** (targeted web search + direct
site/source verification ? NOT the heavy automated workflow, which failed). **Confidence is**
labelled per row. **No pricing is listed because none could be verified** ? every vendor in
this space quotes per-project, so ?/clip is an offline-conversation item (see §6.3). Treat
single-source rows as **leads to verify on the call**, not vetted recommendations.

136

137 ### 6.1 Tier-1 candidates ? verified India presence + video capability (talk to these
first)

138

139 | Vendor | Location (verified) | Video / temporal labeling | Notes for the call |
Confidence |

140 |---|---|---|---|---|

141 | **iMerit** | **Bengaluru** office (A101 & A118, 43 Residency Rd, 560025; tech hub
opened Oct 2022); also Kolkata HQ + US | Yes ? image/video/text/audio; explicit Sports &
Biomechanics domain on their site | Largest & most enterprise-ready (5,500+ annotators); has a
real **Bangalore** footprint for in-person coordination; ask for a sports/temporal-segmentation
reference + a paid pilot. Higher-touch ? likely higher cost. | **High** |

142 | **TELUS International AI** (ex-**Playment**) | Playment was **Bangalore-founded**
(2015); acquired by TELUS 2021, team retained | Yes ? built on Playment's 2D/3D
image+video+LiDAR CV pipeline | Strong CV/video pedigree, Bangalore roots; but now a large
global BPO ? confirm they'll take a **small startup pilot** and who the India delivery contact
is. | **High** (origin) / **Med** (current India ops shape) |

143 | **Taskmonk** | **Bengaluru** | Yes ? "Video" listed as a labeling modality; platform
managed-service workforce | Mid-size, India-local, does both tooling and service ?
potentially the best **cost/agility** fit for a startup pilot. Verify temporal/event labeling
specifically (their heritage is e-commerce catalog tagging). | **Med-High** |

144

145 ### 6.2 Tier-2 ? India video-annotation services, location/specialts to confirm on
contact

146

147 | Vendor | Stated location | Note | Confidence |

148 |---|---|---|---|

149 | **Infolks** | India (Kerala-based per third parties) | Targets startups/SMBs;
image/video/text/audio; positions on cost-efficiency ? good cheap-pilot candidate | **Med** |

150 | **Analytics** | India + USA (since 2018) | General data-annotation incl. video; widely
listed | **Med** |

151 | **Objectways** | HQ Scottsdale AZ; **India delivery in Coimbatore** (not Bangalore) |
Does video/LiDAR; India delivery but not Bangalore-local | **Med** |

152 | **Roora** (Roora ML Pvt Ltd) | **Bengaluru** ? HSR Layout (verified, `rooragrp.com`) |
Image/video/text annotation; sports industry listed. **Selected for the initial paid pilot**
(2026-06-02) ? existence/capability/location verified; **quality TBD via the pilot**. |
Verified (capability) · pilot pending |

153 | **Tika Data** / **Zuru** | Claimed **Bangalore** (single-source listicle each) |
Smaller/startup annotation shops ? unverified beyond one directory mention; verify they exist &
do temporal video before investing time | **Low** |

154

155 ### 6.3 Explicitly NOT verified / out of scope

156 - **Pricing for everyone** ? all quote-only; do not assume. Get per-video-minute (or
per-rally) quotes during the offline conversations.

157 - **Karya (Bengaluru)** ? confirmed local and reputable, but **speech/text/translation**
only, no video ? **excluded** for our use case.

158 - **Sports/temporal track record** ? beyond iMerit's stated "Sports & Biomechanics"
domain, vendors' specific **rally-boundary / temporal-event** experience is unconfirmed marketing
copy; probe with the pilot (§3).

159 - **DPDP / DPA / ISO posture** ? not verified per-vendor; make it a checklist item in
the bake-off (§3 Step 2 security row).

160

161 ### 6.4 Recommendation

162 For the **offline conversations**, start with **iMerit** (Bangalore-local, enterprise QA, sports domain) and **Taskmonk** (Bangalore-local, likely more startup-friendly cost/agility) as the two primary contacts, with **TELUS/Playment** as a third if you want a heavyweight CV pedigree. Run the §3 bake-off across whichever 2/3 agree to a paid pilot, and decide on measured quality-per-rupee. **Verify every "Confidence: Med/Low" claim on the call before committing.**

163

164 > **Update (2026-06-02):** the founder opted to start with **Roora** (Bangalore-local, startup-friendly) as a **single-vendor pilot** rather than the iMerit/Taskmonk multi-vendor bake-off above. iMerit / Taskmonk / TELUS remain the fallback shortlist if the Roora pilot doesn't clear the §3/§4 quality bar.

165

166 > Sources (manual pass, 2026-05-30): iMerit Bengaluru hub ? imerit.net press / PRNewswire (Oct 2022) + contact listings; Playment/TELUS ? BusinessWire & Inc42 (2021 acquisition, Bangalore-founded); Taskmonk & Karya locations ? company sites + AnalyticsIndiaMag / IndiaAI; others ? AnalyticsIndiaMag / Ziloservices / vendor sites. Pricing intentionally omitted (unverifiable).

167

168 ---

169

170 ## 7. Suggested phasing

171

Phase	Deliverable	Depends on
0	Annotation guideline + internal gold key (5?10 clips)	nothing ? can start now
1	Vendor bake-off ? pick winner(s)	§6 research + guideline
2	`AnnotationProvider` registry + `FileProvider` + `HumanVendorProvider`	winner onboarded
3	`AutomatedProvider` (independent oracle) + `regress()` + baseline snapshots	Phase 2
4	Wire surprise triggers (CI / on-demand / scheduled)	Phase 3

179

180 Phase 0 is unblocked today and is the highest-leverage step regardless of vendor choice.

181

182 > **Status (2026-06-02):** Phase 0 guideline ? authored (collector `docs/annotation/`); internal gold key still pending. Phase 1 is running as a **single-vendor Roora pilot** across **3 sports** (badminton, pickleball, table tennis) rather than a multi-vendor bake-off. Phase 2's `FileProvider` (GS-1) and Phase 3's `regress()` + automated oracle (GS-2/GS-4) have already landed; the main remaining slice is the **vendor-dependent `HumanVendorProvider`** (GS-3), with trigger-wiring (GS-5) after it.

183

184 ---

185

186 ## 8. Open questions

187 1. **Guideline authorship** ? who owns the canonical rally definition, and is one expert's key enough, or do we want 2-expert consensus for the answer key itself?

188 2. **Corpus representativeness** ? how well does the owned/licensed collection mirror eventual user capture? (Ties directly to VIDEO_RECORDING_GUIDELINES.md.)

189 3. **Baseline drift** ? when production legitimately improves, how/when do we re-snapshot `baseline.json` so old baselines don't raise false regressions?

190 4. **Fast-tier oracle cost** ? per-video Gemini cost at regression frequency (see `archives/MONETIZATION\_AUDIT.md` §4 for the per-video estimate).

191

9. user (2026-06-12T09:03:33.494Z)

1 # Golden-Set Subproject ? Design & Implementation Overview

2

3 **Status:** In progress . **Date:** 2026-05-30

4 **Companion docs:** design rationale + vendor shortlist in GOLDEN_SET_VENDORING.md (ADR-006); scoring mechanics in EVALUATION.md.

5

6 This is the **implementation tracker** for turning the golden-set/vendor **design** into shipped code. It's split into small, independently-reviewable PRs. The whole subproject is **vendor-agnostic by construction** ? only one slice (GS-3, `HumanVendorProvider`) depends on

which vendor you pick offline; everything else can land first.

```
7
8   > **Update (2026-06-02):** Vendor selected for the **initial pilot ? Roora**
(Bengaluru). The **annotation guideline, intake-manifest template, and NDA** are authored in the
collector repo `docs/annotation/`, and the collector's `import-local` ingestion now covers the
**3 pilot sports** (badminton, pickleball, table tennis). On the indexer side, GS-1
(`file_provider.py`), GS-2 (`backend/eval/regression.py` + `run_regression.py`), and the GS-4
oracle scaffold (`automated_provider.py`) have landed; the remaining slice is the vendor-
dependent **GS-3 `HumanVendorProvider`**, parked until the Roora pilot confirms quality.
9
10    ---
11
12    ## Architecture in one picture
13
14    ```
15          video_ref ??? AnnotationProvider.submit()/poll()/fetch() ???
List[(start,end)]  ground truth
16
17          ?           ?           ?           ?
18          FileProvider HumanVendor Automated(oracle)
19          (GS-1)       (GS-3)       (GS-4)
20 backend.eval.metrics.score() ??? existing, unchanged
21
22          production pipeline ??? DB ??? backend.eval.harness.get_predictions() ???
List[(start,end)]  predictions
23
24          ?
25          regress(video, tier) (GS-2)
26 compares the two
27    ```
28
29    **The keystone:** a provider's `fetch()` returns the **same `List[(start, end)]`** that
`eval.annotations.load_annotations()` already returns and `eval.metrics.score()` already
consumes. So one scorer grades all three of: product-vs-GT, vendor-vs-answer-key, and annotator-
vs-annotator (IAA). No new scoring code ? only new *sources* of intervals.
30
31    ---
32
33    ## PR breakdown
34
35    | Slice | Branch | Scope | Depends on | Vendor-dependent? |
36    |---|---|---|---|---|
37    | **GS-1** | `feat/annotation-provider-registry` | `AnnotationProvider` ABC + registry +
`FileProvider` + tests | ? | No |
38    | **GS-2** | `feat/regression-runner` | `regress(video, tier, tolerance)` over existing
harness + `baseline.json` snapshot/compare + CLI | GS-1 | No |
39    | **GS-3** | `feat/human-vendor-provider` | `HumanVendorProvider` (submit/poll/fetch
against the chosen vendor's API or file-drop), DPA/secure-link handling | GS-1, **your vendor
pick** | **Yes** |
40    | **GS-4** | `feat/automated-oracle-provider` | `AutomatedProvider` fast-tier oracle ?
**interface stub** (injected labeler + fake; concrete model = backlog) | GS-1, GS-2 | No |
41    | **GS-5** | `feat/regression-triggers` | Wire triggers: CI gate, on-demand CLI,
scheduled run | GS-2, GS-4 | No |
42
43    Each row is one PR. GS-1 ? GS-2 ? GS-4 ? GS-5 can proceed **now**, independent of vendor
selection. GS-3 is parked until you've chosen a vendor (then it's a thin adapter behind the GS-1
interface).
44
45    ---
46
47    ## Slice details
48
49    ### GS-1 ? Provider registry + FileProvider ? (this PR)
50    - `backend/annotations/base.py` ? `AnnotationProvider` ABC (`submit`/`poll`/`fetch` +
`annotate`) convenience) and `annotation_registry`, mirroring `backend/providers` and
`backend/indexer` registry patterns.
51    - `backend/annotations/file_provider.py` ? `FileProvider`: the "job" is a CSV that
```

```

already exists (formalizes today's manual/ShuttleSet flow); reuses
`eval.annotations.load_annotations`.
48     - Tests assert the registry contract **and** the keystone (provider output is directly
scorable by `metrics.score()`).
49     - **Design choice:** `fetch()`/`annotate()` raise loudly on a missing/incomplete job
rather than returning `[]`, so a broken label source can't silently masquerade as "0 rallies"
and skew a regression to green.
50
51     ### GS-2 ? Regression runner (next)
52     `regress(video, tier="fast", tolerance=0.05)`: get GT from
`annotation_registry.get(tier)`, read predictions via the existing `eval.harness`, score with
the existing `metrics`, compare against a per-video `baseline.json`, and flag a regression when
precision/recall drop beyond tolerance. Pure orchestration over existing parts + a small
baseline store. CLI: `python run_regression.py --video ... --tier file`.
53
54     ### GS-3 ? HumanVendorProvider (after your vendor pick)
55     A concrete provider that submits a (licensed/owned-only) video to the chosen vendor and
fetches the returned rally CSV. Carries the privacy controls from the design doc (secure time-
limited links, delete-on-delivery, DPA). Thin adapter ? the interface already exists from GS-1.
56
57     ### GS-4 ? AutomatedProvider (fast-tier oracle) ?? oracle rule
58     An independent strong labeler for the fast/CI tier. **Must be a different model lineage
than whatever production uses** (else shared blind spots ? false green). A fast-tier failure is
a *smoke alarm* that escalates to the GS-3 human tier, never an authoritative verdict on its
own.
59
60     ### GS-5 ? Regression triggers
61     Wire `regress()` into CI (pre-merge gate on pipeline changes), an on-demand CLI (the
"surprise regression"), and an optional scheduled run. Includes a deliberate re-snapshot path so
legitimate production improvements can rebaseline instead of raising false regressions.
62
63     ---
64
65     ## Conventions
66     - **Branch per slice**, one PR each, tests green before PR, full suite must stay green.
67     - New runtime deps: **none** through GS-2 (GS-3 may add a vendor SDK; GS-4 may reuse the
existing `google-genai` for a Gemini oracle).
68     - All ground-truth video stays **licensed/owned** ? see the privacy section of the
design doc.
69

```

10. user (2026-06-12T09:03:33.740Z)

```

1     # Phase-1 Golden-Set Video Collection Spec
2
3     **Status:** Design v1 . **Date:** 2026-06-02
4     **Companion docs:** [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md) (vendor process,
ADR-006) . [VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md) (capture spec the
user-facing guide derives from) . scoring in [EVALUATION.md](EVALUATION.md).
5
6     What footage to gather for the **first round** of golden-set creation so the trained
rally segmenter reaches good precision/recall ? and so the eval number actually predicts
production. Golden-set creation is **iterative**: expect multiple rounds, expanding toward
whatever axis round-1 metrics show is weakest.
7
8     ---
9
10    ## 1. The principle that dominates everything: match the deployment distribution
11
12    The CV literature is consistent: **models trained on broadcast footage fail to
generalize to self-recorded amateur video** ? broadcast uses a "straight, fixed camera with
minimum variability," while real non-elite footage has limited coverage, occlusions, and
inconsistent viewpoints ([team-sports
survey](https://link.springer.com/article/10.1007/s10462-024-10934-9), [unstructured-video event
detection](https://arxiv.org/pdf/2002.08097)). This is the **same failure mode as ADR-004**

```

```

(broadcast cameras *track players*, so player pixel-motion inverts during rallies).
13
14     Our product ingests **users' own match recordings** (a fixed phone/GoPro on a tripod).
Therefore:
15
16     - The golden set ? **especially the held-out eval split** ? must be dominated by
**static, self-recorded, full-court** footage that mirrors real users.
17     - **ShuttleSet / broadcast** footage is fine as *training* augmentation and as a
contrast set, but must be **minimal in eval**. Measuring precision/recall on broadcast yields a
number that does **not** predict production.
18
19     ---
20
21     ## 2. Primary recording profile (also the basis of the user-facing capture guide)
22
23     Grounded in the badminton-CV setup literature (camera ~2 m behind the baseline, ~4.5 m
elevated, ~30° down-tilt, whole court + headroom above the net ?
[source](https://pmc.ncbi.nlm.nih.gov/articles/PMC10747833/)):
24
25     | Setting | Spec | Why |
26     |---|---|---|
27     | Mount | Fixed tripod; **no pan/zoom** during play | Avoids the ADR-004 camera-tracking
trap |
28     | Position | Behind one baseline/end, centered | Full-court, symmetric view |
29     | Height | **375 m** elevated (~4.5 m ideal) | De-occludes the far player & net |
30     | Tilt | **~30° down** | Optimal coverage, low keystoneing |
31     | Framing | Entire court **+ headroom above the net**; players never leave frame |
Detection + court-relevance need it |
32     | Lighting | Even; no backlight/glare | Reduces blur & false motion |
33
34     Most of round 1 should use this profile; a minority deliberately varies it (§4) to probe
robustness.
35
36     ---
37
38     ## 3. Resolution & frame rate (per sport)
39
40     A controlled study: table-tennis ball detection was **22.5% @ 30 fps ? 93.2% @ 60 fps**
(120 fps ? 60 fps) ([Nature Sci Rep](https://www.nature.com/articles/s41598-023-42966-6));
OpenTTGames records at 120 fps. The ball/shuttle is the camera-invariant rally signal (our WASB
substrate, ADR-001), so fps is **not** a nicety for fast sports.
41
42     | Sport | Min resolution | Recommended fps | Note |
43     |---|---|---|---|
44     | Badminton | 1080p (720p floor) | 30 ok, **50/60 better** | Shuttle decelerates fast;
60 sharpens smashes |
45     | Pickleball | 1080p | **30?60** | Ball slower than TT; 30 likely fine, 60 safer |
46     | Table tennis | 1080p | **60 min (120 ideal)** | 30 fps collapses ball detection |
47
48     > These are **literature-backed starting points**; the published minimums get finalized
empirically against our golden set (VIDEO_RECORDING_GUIDELINES.md §3).
49
50     ---
51
52     ## 4. The Phase-1 diversity matrix
53
54     Cover the **high-impact axes** now; don't chase rare conditions in round 1. "Guards" =
whether the axis mostly protects **P**recision (not calling dead-time a rally) or **R**ecall
(not missing rallies).
55
56     | Dimension | Vary across? | Guards | Round-1 target |
57     |---|---|---|---|
58     | **Sport** | badminton, pickleball, table tennis | both | all 3 |
59     | **Format** | singles **and** doubles (badminton, pickleball) | R | both for those 2 |
60     | **Camera profile** | mostly primary-static; 1?2 side-elevated; 1?2 lower/phone-height;

```

```

**1?2 broadcast (contrast only)** | both | ~80% static / ~20% variants |
61 | **Venue / court color** | wood, green, blue synthetic | P | ?3 venues |
62 | **Lighting** | bright gym, dim club, mixed/backlit | both | ?3 conditions |
63 | **Background clutter** | empty hall vs people moving behind court | **P** (crowd
motion ? false positives) | both |
64 | **Players** | skill, attire colors, body types, L/R-handed, skin tone/age | R |
spread; don't reuse one pair |
65 | **Rally edge cases** | service faults, lets/replays, 1?2-shot & long rallies, net-
cords, coaching pauses, warm-up | boundary **P/R** | ensure present |
66 | **Resolution/fps ablation** | same match re-encoded 480/720/1080p; 30 vs 60 fps |
calibration | 1 matched set/sport |
67
68 ---
69
70 ## 5. Split it so the precision/recall number is honest (no train/eval leakage)
71
72 The #1 video-ML mistake: the **same match in train and eval inflates the score** ([data-
leakage guide](https://towardsdatascience.com/data-science-mistakes-to-avoid-data-
leakage-e447f88aae1c/)).
73
74 - **Split by match / recording session ? never by clip.** All clips from one match go
entirely to train *or* eval.
75 - **Hold out whole *conditions*** (a venue, a camera profile, a player pair the model
never saw) so eval measures generalization, not memorization.
76 - Reserve **~? of each sport as a deployment-like held-out eval set** (static, self-
recorded). Broadcast/ShuttleSet may enter train but should be **minimal in eval**.
77
78 > **Guardrails ? TODO (design with the user):** train and eval live in **separate
catalogs** today, kept apart by convention. We want hard guardrails ? permissions / explicit
approval / re-confirmation at ingest & export ? so a video can **never** silently land in both.
Tracked in BACKLOG.md; to be designed in a later session.
79
80 ---
81
82 ## 6. Concrete "round 1" shopping list (per sport)
83
84 Aligns with the 12?15-videos/sport scale plan; the 3-clip Roora calibration is a subset
of this.
85
86 - **8 static-primary** clips spanning venues / lighting / skill (?4 singles, 4 doubles
where applicable)
87 - **2 camera-variant** clips (side-elevated, lower height) ? robustness probe
88 - **1 resolution/fps ablation set** (one match @ 480/720/1080p, 30 & 60 fps)
89 - **1 broadcast contrast** clip (ShuttleSet) ? *train/contrast only*
90 - Each ~2?4 min, **deliberately including the edge cases** above ? **~12 clips/sport** ;
hold out **4 whole matches** for eval.
91
92 ---
93
94 ## 7. Deliberately defer to later rounds
95
96 Outdoor courts, phone-handheld/jitter, extreme angles, 4K, unusual net/court variants,
crowd-heavy tournaments. Add these once round-1 precision/recall shows the weakest axis, then
expand toward it (find the quality "knee" ? VIDEO_RECORDING_GUIDELINES.md §3).
97
98 ---
99
100 ## 8. Two cautions specific to precision (the stated goal)
101
102 - **Backgrounds with people moving** behind the court are the biggest false-positive
driver. Include some so the model learns to reject them ? and make sure the annotation
guideline's "ignore all dead time" rule is airtight.
103 - **Warm-up / knock-up / practice swings** look like rallies. Capture some, and confirm
the vendor **excludes** them (the Roora guide now has an explicit rule + a how-to-spot-it tip).
Early precision is bounded more by **guideline / inter-annotator consistency (IAA)** than by raw

```

data volume.
104
105 ---
106
107 ## 9. Sources
108
109 - [Team-sports action-recognition survey ? Springer
2024](https://link.springer.com/article/10.1007/s10462-024-10934-9)
110 - [Unstructured-video event detection ? arXiv
2002.08097](https://arxiv.org/pdf/2002.08097)
111 - [Badminton service-height CV setup (camera 4.5 m / 30°) ?
PMC](https://pmc.ncbi.nlm.nih.gov/articles/PMC10747833/)
112 - [VideoBadminton dataset ? arXiv 2403.12385](https://arxiv.org/html/2403.12385v1)
113 - [Table-tennis frame-rate vs detection accuracy ? Nature Sci
Rep](https://www.nature.com/articles/s41598-023-42966-6)
114 - [OpenTTGames (120 fps) dataset](https://lab.osai.ai/)
115 - [Action-spotting / segmentation survey ? arXiv
2505.03991](https://arxiv.org/html/2505.03991v2)
116 - [Data-leakage in train/test splits](https://towardsdatascience.com/data-science-
mistakes-to-avoid-data-leakage-e447f88aae1c/)
117

11. user (2026-06-12T09:03:34.562Z)

Found 37 files
docs\NEXT_STEPS.md
docs\BACKLOG.md
docs\CODE_MAP.md
docs\DEFERRED_HARDENING_PLAN.md
scratch\PR2_HANDOFF.md
docs\archives\TEST_RUN_SUMMARY.md
README.md
docs\archives\decisions\DECISIONS.md
docs\UI_PROPOSAL.md
docs\HOSTING_PLAN.md
scratch\PR1_HANDOFF.md
docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\COMMERCIALIZATION.md
docs\GOLDEN_SET_VENDORING.md
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs\PLATFORM_ARCHITECTURE.md
docs\ROADMAP.md
CLAUDE.md
scratch\HIGH_RISK_REMEDIATION_PLAN.md
docs\archives\quality-iterations\README.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\VIDEO_RECORDING_GUIDELINES.md
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\archives\MONETIZATION_AUDIT.md
docs\PMF_MARKET_RESEARCH.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\GOLDEN_SET_IMPLEMENTATION.md
docs\EVALUATION.md

12. assistant (2026-06-12T09:03:36.667Z)

Now let me search for more specific mentions of delays, offsets, measured numbers:

13. user (2026-06-12T09:03:37.508Z)

Found 82 files

```
tests\test_upload_download.py
backend\config\models.py
backend\pipeline\detectors\base.py
backend\pipeline\detectors\tracknet_runner.py
tests\test_wsl_tilde_quoting.py
tests\test_trajectory_hybrid_equivalence.py
tests\test_tracknet_hybrid.py
tests\test_config.py
backend\eval\calibrate_local.py
backend\eval\distill_local.py
backend\eval\harness.py
backend\pipeline\audio\hit_detector.py
backend\pipeline\detectors\wasb_infer.py
backend\pipeline\validators\base.py
backend\utils\paths.py
tests\test_wasb_cache.py
tests\test_yolo_hybrid.py
training\gen0\harness.py
tests\test_training_gen0.py
tests\test_reporting.py
tests\test_rally_seq_proto.py
tests\test_provider_gemini.py
tests\test_path_safety.py
tests\test_geometry.py
tests\test_gemini.py
tests\test_eval_regression.py
tests\test_database.py
tests\test_eval_classifier.py
tests\test_base.py
run_process_and_stitch.py
backend\reporting\__init__.py
backend\pipeline\validators\scene_cut.py
backend\pipeline\validators\pts_continuity.py
backend\pipeline\validators\blur.py
backend\pipeline\stitcher\compiler.py
backend\pipeline\segmenters\yolo_hybrid.py
backend\pipeline\segmenters\motion.py
backend\pipeline\segmenters\gemini.py
backend\pipeline\segmenters\base.py
backend\eval\rally_seq_proto.py
backend\eval\eval_partial_labels.py
backend\eval\calibration.py
backend\eval\calibrate_wasb.py
backend\providers\gemini.py
tests\test_run_signals.py
backend\reporting\harvest.py
tests\test_reporting_harvest.py
tests\test_import_smoke.py
run_regression.py
tests\test_eval_metrics.py
backend\eval\metrics.py
backend\eval\regression.py
tests\test_reingest_safety.py
tests\test_detector_keying.py
tests\test_eval_partial_labels.py
tests\test_tracknet_runner.py
run_eval.py
run_phase3_eval.py
backend\tools\export.py
backend\eval\batch.py
```

```
tests\test_hit_detector.py
tests\test_eval_harness.py
backend\pipeline\detectors\rally_gate.py
backend\eval\export_wasb_segments.py
backend\eval\audio\e1_probe.py
backend\eval\gemini_refine.py
tests\test_export_wasb_segments.py
tests\test_gemini_refine.py
tests\test_calibrate_wasb.py
tests\test_calibration.py
backend\tools\rally_slicer.py
tests\test_annotation_providers.py
tests\test_eval_training.py
scratch\test_thresholds.py
scratch\gemini_video_debugger.py
scratch\diagnose_rally.py
scratch\analyze_motion.py
backend\utils\geometry.py
backend\sports\badminton.py
backend\eval\training.py
backend\eval\classifier.py
backend\eval\__init__.py
```

14. user (2026-06-12T09:03:38.129Z)

```
docs\BACKLOG.md:47:- ? **Vendor pilot (Roora)** ? run the paid pilot with **Roora** as a
**single-vendor pilot first** (not a multi-vendor bake-off): assess quality-per-rupee against
the $4 bar; escalate to the iMerit/Taskmonk fallback only if it misses. **Revisit:** the pilot
doubles as calibration once the gold key exists.
docs\GOLDEN_SET_VENDORING.md:10:[Omitted long matching line]
docs\GOLDEN_SET_IMPLEMENTATION.md:8:[Omitted long matching line]
```

15. user (2026-06-12T09:03:38.840Z)

```
2c241a8 refactor: collapse the hybrid-segmenter twins into a shared trajectory base
62fb2fb docs: sync to ratified decisions + repo reality (audit truth-sync) (#71)
a5477c2 docs: next-steps + multisport findings (WASB transfers to table tennis, not pickleball)
(#62)
26ee2de docs: next-steps + multisport findings (WASB transfers to table tennis, not pickleball)
d05fab0 docs: merge monetization reviews into a single living COMMERCIALIZATION tracker
59e813d chore: remove vendored VLC annotator (lives in its own repo now) (#33)
77b6083 docs: ending_reason is now a forced/unforced-error taxonomy
35be998 Merge pull request #17 from avidullu/docs/golden-set-status-update
8e8c5c2 docs(backlog): mark annotation guideline done; Roora pilot replaces bake-off
36044c5 docs: sync golden-set + ADRs after Roora pilot + multi-sport ingestion
1ce2425 feat(eval): regression runner + baseline store + CLI (GS-2)
3e22cb4 feat(annotations): AnnotationProvider registry + FileProvider (GS-1)
14422e0 Merge pull request #12 from avidullu/docs/vendor-shortlist
7e5add6 docs(golden-set): add verified Bangalore/India vendor shortlist ($6)
f5f50d8 Merge pull request #10 from avidullu/docs/golden-set-vendoring
f2ad8cd docs: golden-set generation + vendor process (v1), recording guidelines, ADR-006
db3d92e Add chunked processing for long videos: split into overlapping segments, process each
via Gemini independently, offset and merge results
```

16. assistant (2026-06-12T09:03:40.596Z)

Let me check the EVALUATION.md and look into the code more carefully:

17. user (2026-06-12T09:03:46.058Z)

```
1      # Annotation & Evaluation
2
3      How to measure how well the pipeline detects rallies, using hand-annotated
4      ground-truth timestamps. The harness scores predictions **already stored in the
```

```

5 SQLite DB** ? it makes no API calls and is fully deterministic, so you can run it
6 as often as you like for free.
7
8 ---
9
10 ## 1. What you annotate (and what it does / doesn't do)
11
12 You annotate, per video, the `[start, end)` time range of each **rally**. That's
13 all the evaluation harness needs.
14
15 > **Important:** these rally timestamps measure and tune **rally segmentation**
16 > ? i.e. *where* the pipeline says rallies begin and end. They do **not** train
17 > the shuttle detector (TrackNet/WASB), which learns from per-frame shuttle
18 > `(x, y)` coordinate labels, a different and much heavier annotation. Use rally
19 > timestamps as your **golden evaluation set** ; only invest in frame-level
20 > coordinate labels if evaluation shows *detection* (not segmentation) is the
21 > bottleneck. See `docs/TRACKNET_WSL_SETUP.md`.
22
23 ## 2. Annotation file format (CSV)
24
25 One CSV per video, one rally per row:
26
27 ```csv
28 start,end
29 00:01:12,00:01:39
30 102.5,131.0
31 ```
32
33 - The `start,end` header is optional (auto-detected).
34 - Timestamps may be **decimal seconds** (`102.5`) or **colon-separated**
35   `MM:SS` / `HH:MM:SS` (`00:01:12`). Forms can be mixed.
36 - Blank lines and `#` comments are ignored.
37 - Malformed rows (bad timestamp, `start >= end`) fail loudly rather than being
38   silently skipped ? so labeling mistakes surface immediately.
39
40 Generate an empty template to fill in:
41
42 ```bash
43 python run_eval.py --scaffold --annotations rallies.csv
44 ```
45
46 ## 3. Running the evaluation
47
48 First make sure the video has been processed so its detections are in the DB
49 (the pipeline keys each video by its file MD5). Then:
50
51 ```bash
52 # Identify the video by file (its MD5 is computed for you)...
53 python run_eval.py --video "C:/videos/match.mp4" --annotations rallies.csv
54
55 # ...or by the stored video_id (MD5) directly:
56 python run_eval.py --video-id <md5> --annotations rallies.csv --stage both --iou 0.5
57 ```
58
59 Useful flags:
60
61 | Flag | Effect |
62 |-----|-----|
63 | `--stage {candidates,final,both}` | Which prediction stage(s) to score (default
64 `both`). |
65 | `--detector <name>` | Only score candidates from one detector (e.g. `yolo_hybrid`,
66 `tracknet_hybrid`). |
67 | `--iou <float>` | Match threshold (default `0.5`). |
68 | `--pr-curve` | Also report an IoU sweep (0.1?0.9). |
69 | `--show-failures` | List missed (FN) and spurious (FP) rallies with timestamps. |

```



```

68 | `--json <path>` | Write a machine-readable report. |
69 | `--db <path>` | Override the DB path (default: `output/<output.db_name>` from
`config.json`). |
70
71 ## 4. Reading the output
72
73 ```
74 =====
75 Rally evaluation ? video 'demo'
76 Ground-truth rallies: 3 | IoU threshold: 0.5
77 =====
78 Stage      Pred  TP  FP  FN  Prec  Rec  F1  mIoU  dStart  dEnd
79 -----
80 candidates    3   3   0   0  1.00  1.00  1.00  1.00  0.0s   0.0s
81 final          2   2   0   1  1.00  0.67  0.80  0.95  0.2s   0.5s
82 =====
83 [final] missed rallies (false negatives): 1
84   - 00:01:20?00:01:32
85 ```
86
87 - **TP / FP / FN** ? true positives (matched), false positives (spurious
88   predictions), false negatives (missed rallies).
89 - **Prec / Rec / F1** ? precision, recall, and their harmonic mean.
90 - **mIoU** ? mean temporal IoU over matched pairs (how tightly matched rallies
91   overlap their ground truth).
92 - **dStart / dEnd** ? mean absolute boundary error in seconds (are rallies
93   detected but trimmed in the wrong place?).
94
95 ### The key diagnostic: candidates vs. final
96
97 Scoring both stages tells you where to spend effort:
98
99 - **Candidates miss rallies the GT has** ? the **detector** is the bottleneck.
100   Improve the shuttle model (e.g. fine-tune TrackNet/WASB with frame-level
101   labels) or relax detection thresholds.
102 - **Candidates catch them but `final` is wrong/trimmed** ? the **segmentation /
103   AI handoff** is the bottleneck. Tune thresholds (`velocity_threshold`,
104   `dead_time_merge_gap`, `min_action_window_duration`) or the AI prompt ? no new
105   labels needed.
106
107 In the example above, detection is perfect (candidates: recall 1.0) but the AI
108 handoff dropped one rally (final: recall 0.67) ? so the fix is in segmentation,
109 not detection.
110
111 ## 5. How matching works
112
113 Predictions and ground truth are sets of `[start, end)` intervals. A prediction
114 **matches** a ground-truth rally when their **temporal IoU** (overlap ÷ union) is
115 at least the threshold. Matching is **greedy and one-to-one** ? highest-IoU pairs
116 are claimed first, and each prediction/GT is used at most once, so a single
117 detection can't be credited for two rallies. Unmatched predictions are false
118 positives; unmatched ground-truth rallies are false negatives.
119
120 ## 6. Code map
121
122 | File | Role |
123 |-----|-----|
124 | `backend/eval/metrics.py` | Pure interval math: IoU, matching, P/R/F1, boundary error.
125 |
126 | `backend/eval/annotations.py` | Load/validate the ground-truth CSV. |
127 | `backend/eval/harness.py` | Read predictions from the DB, score, format the report. |
128 | `run_eval.py` | CLI entry point. |
129 | `tests/test_eval_*.py` | Deterministic tests for all of the above. |
130
131 ## 7. Troubleshooting

```

```

131
132     | Symptom | Cause & fix |
133     |-----|-----|
134     | `ERROR: database not found at output/...` | The harness only reads stored predictions
? it does not run the pipeline. Process the video first (e.g.
`run_process_and_stitch.py`) so detections land in the DB, then re-run. |
135     | Every rally shows as a miss (FN), `Pred` is 0 | This video hasn't been processed, or
you used the wrong `video_id`. The pipeline keys videos by file MD5 ? pass the same file
via `--video`, or the exact stored `--video-id`. |
136     | `video_id` doesn't match although it's "the same" video | Re-encoding/trimming changes
the bytes, hence the MD5. Use the exact file that was processed, or look up the stored id (it's
printed during processing). |
137     | `candidates` stage is empty but `final` has data (or vice-versa) | Depends on which
segmenter ran. `gemini` writes only `final` (no candidates); `yolo_hybrid`/`tracknet_hybrid`
write both. Use `--stage final` for the gemini segmenter, and `--detector <name>` to scope
candidates. |
138     | Recall looks low only at high `--iou` | Rallies are found but boundaries are loose ?
check `dStart`/`dEnd`. Tune `stitching` padding or the segmentation thresholds rather than the
detector. |
139     | `ValueError: start (...) >= end (...)` while loading annotations | A bad row in your
CSV. Timestamps must be increasing per rally and use seconds or `HH:MM:SS`. |
140
141     Run `python run_eval.py -h` for a quick reminder of this prerequisite and all flags.
142

```

18. user (2026-06-12T09:03:46.300Z)

```

Found 15 files
backend\eval\calibrate_local.py
backend\eval\distill_local.py
backend\eval\harness.py
backend\eval\rally_seq_proto.py
backend\eval\eval_partial_labels.py
backend\eval\calibration.py
backend\eval\calibrate_wasb.py
backend\eval\metrics.py
backend\eval\regression.py
backend\eval\export_wasb_segments.py
backend\eval\audio\e1_probe.py
backend\eval\gemini_refine.py
backend\eval\training.py
backend\eval\classifier.py
backend\eval\__init__.py

```

19. user (2026-06-12T09:03:46.561Z)

```

Exit code 1
/usr/bin/bash: line 1: $null: ambiguous redirect

```

20. assistant (2026-06-12T09:03:48.802Z)

Let me search in the sports-data-collector repo for Roora-specific info:

21. user (2026-06-12T09:03:49.345Z)

```

C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ANNOTATION_GUIDE.md:> from the
pilot videos before sending this guide to Roora.**
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\NDA_TEMPLATE.md:> `TODO(avidullu)`
blank, then send to Roora for signature before any footage is shared.
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\NDA_TEMPLATE.md:-
Disclosing/Receiving Party B: Roora ML Pvt Ltd, 27th Main, 13th Cross, HSR Layout Sector 1,
Bangalore, Karnataka, India ("Party B").
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\NDA_TEMPLATE.md:Party B (Roora
ML Pvt Ltd)

```

```

C:\Users\avidu\Projects\sports-data-collector\docs\annotation\NDA_TEMPLATE.md:Name:
**TODO(avidullu): Roora signatory name**
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ROORA_BRIEF.md:# Roora Annotation
Brief ? Rally Golden Set
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ROORA_BRIEF.md:- **TODO(avidullu):
paste the Google Drive folder link** where source videos are shared with Roora.
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ROORA_BRIEF.md:- **TODO(avidullu):
paste the Drive folder link** where Roora uploads completed `

```

22. assistant (2026-06-12T09:03:51.430Z)

Let me read the Roora brief and annotation guide:

23. user (2026-06-12T09:03:51.555Z)

```

1      # Roora Annotation Brief ? Rally Golden Set
2
3      **Project:** racket-sport rally segmentation golden / regression dataset
4      **Sports in scope:** Badminton, Pickleball, Table Tennis (each in **singles _and_
doubles**)
5      **Companion doc:** [ANNOTATION_GUIDE.md](./ANNOTATION_GUIDE.md)
6
7      > This brief is the vendor-facing spec. The annotation deliverable is a per-rally CSV
that
8      > imports directly via `python -m src.cli.curate import-local`.
9
10     ## 0. What we need in one sentence
11     For each provided match video, produce **one CSV file** listing **every rally** with its
12     start time, end time, number of shots, and how it ended ? **excluding all dead time**.
13
14     ## 1. Deliverable format (canonical)
15     One CSV per video (UTF-8), named exactly `

```

```

35     ``
36     =IF(B2="", "", IFERROR(VALUE(LEFT(B2,FIND(":",B2)-1))*60 +
VALUE(MID(B2,FIND(":",B2)+1,20)), VALUE(B2)))
37     ``
38
39     On **Download as CSV**, the computed decimal seconds are exported. (If your tool already
40     shows decimal seconds, type those into `start_time`/`end_time` and leave the mm:ss
columns blank.)
41
42     Example (badminton):
43     ```csv
44     rally_number,start_mmss,end_mmss,start_time,end_time,shots_count,ending_reason,last_shot
_outcome,score_before,score_after,notes
45     1,0:12.50,0:25.00,12.50,25.00,9,winner,in,0-0,1-0,
46     2,0:45.00,0:47.30,45.00,47.30,1,service_fault,n_a,1-0,1-1,serve into net
47     3,1:01.20,1:18.85,61.20,78.85,14,unforced_error,out,1-1,2-1,final clear sailed long (no
pressure)
48     ``
49
50     **Critical format rules**
51     - `start_time`/`end_time` that reach us **must be decimal seconds** (the formula
guarantees this).
52     - Times are relative to the **start of the file we provide**. Do **not** trim, clip, or
53     re-encode the video ? our pipeline identifies a video by its file checksum.
54     - `end_time` > `start_time` for every row.
55     - No overlaps: `end_time[N]` <= `start_time[N+1]`.
56     - Everything between one rally's `end_time` and the next rally's `start_time` is **dead
time** and gets **no row**.
57
58     ## 2. The three non-negotiable rules (full detail + pictures in the Guide)
59     1. **Ignore all dead time.** Only label active play.
60     2. **Include every rally ? exclude none.** Every served point, including **service
faults**,
61     1?2 shot rallies, and **lets/replays**. A missed rally is the worst error in this
project.
62     **Exception: warm-up / knock-up is NOT a rally** ? pre-match and between-games warm-
up looks
63     like rallying but isn't scored play, so **don't label it**. Tell it apart by: no real
serve,
64     cooperative (not competitive) hitting, the score not progressing, often several
shuttles/balls
65     in use (full how-to in the Guide, Part 2). When unsure if the match has started, flag
it.
66     3. **A shot = one contact** with the racket/paddle/bat. Serve = shot 1. Bounces and net-
clips are not shots.
67
68     ## 3. Scope & phasing
69     **Phase A ? Pilot (this engagement):** 3 videos (1 badminton + 1 pickleball + 1 table
tennis),
70     each `< 3 min` with ~10?12 rallies. This is the **paid calibration round**; we review
jointly and
71     lock the Guide before scale-up.
72
73     **Phase B ? Scale (after a successful pilot):** ~12?15 videos **per sport** (~36?45
total), each
74     ~30 min. Same format, rules, and template.
75
76     > **Doubles are included** across the program. Rally rules are identical to singles; the
77     > badminton- and pickleball-specific doubles notes are in the Guide.
78
79     ## 4. Quality bar (acceptance criteria)
80     Automated validation (no overlaps, positive durations, schema-valid) runs on every file,
then a
81     ~20% human spot-check. A file is returned for rework (in scope) if it misses:
82     - **Completeness:** every rally present; zero missed rallies; zero dead-time rows.

```

```

83 - **Boundaries:** `start_time`/`end_time` within **±0.3 s** of true serve-contact /
landing.
84 - **shots_count:** exact for rallies ? 15 shots; ±1 allowed for longer/faster rallies.
85 - **ending_reason:** one of the six allowed values, per the Guide's decision tree.
86 - **Format:** decimal seconds present, valid columns, non-overlapping, chronological.
87
88 ## 5. Per-sport quick reference (full version + examples in the Guide)
89 **Badminton (singles & doubles)** ? start: racket?shuttle serve contact · end: shuttle
floors / fault · shot: each racket?shuttle contact.
90 **Pickleball (singles & doubles)** ? start: paddle?ball serve contact · end: double
bounce / out / net / fault · shot: each paddle?ball contact (floor bounces are **not** shots;
two-bounce rule is normal play).
91 **Table tennis** ? start: racket?ball serve strike (not the toss) · end: failed legal
return · shot: each racket?ball contact (table bounces are **not** shots). Net-cord serve =
`let`.
92
93 ## 6. ending_reason values (decision tree in the Guide)
94 This vocabulary attributes **why** the point ended; it is shared across all
95 net-separated sports (badminton, pickleball, table tennis, tennis). Record **where**
96 the last shot physically landed in the separate **`last_shot_outcome`** column
97 (`in` \| `net` \| `out` \| `n_a`) ? e.g. a forced error that flew long is
98 `ending_reason=forced_error, last_shot_outcome=out`.
99
100 | value | meaning |
101 |---|---|
102 | `winner` | last shot landed **in** and the opponent made no legal contact (a clean
winner). |
103 | `forced_error` | the losing player erred (netted / out / missed) **under pressure** ?
stretched, wrong-footed, or rushed by the opponent's strong shot. |
104 | `unforced_error` | the losing player erred on a **routine, unpressured** ball they had
time and position to make. |
105 | `service_fault` | the serve itself failed / was illegal. |
106 | `let` | the point was replayed. |
107 | `other` | anything else / cannot tell. **Must** add a note. |
108
109 ## 7. Recommended capture method
110 Use any tool that shows current time (mm:ss.mmm or seconds, or frame + known fps) and
can
111 **step one frame** forward/back: a frame-steppable player (VLC frame-step) or a timeline
tool
112 (Label Studio / CVAT). The deliverable is the CSV in the schema above.
113 `seconds = frame_number / fps` (we provide fps per video).
114
115 ## 8. Logistics & contacts
116 - **TODO(avidullu): paste the Google Drive folder link** where source videos are shared
with Roora.
117 - Per-video `fps` + `video_id`: provided in the intake manifest ? template
[`INTAKE_MANIFEST_TEMPLATE.csv`](./INTAKE_MANIFEST_TEMPLATE.csv). **TODO(avidullu): fill the
manifest (one row per pilot video) and share its link.**
118 - **TODO(avidullu): paste the Drive folder link** where Roora uploads completed
`<video_id>.csv` files.
119 - Blank annotation template Sheet (mm:ss ? seconds formula from $1 pre-filled):
**TODO(avidullu): create it and paste the link.**
120 - **TODO(avidullu): add the escalation channel** (email / WhatsApp group) for edge-case
questions.
121 - NDA before any footage is shared ? template [`NDA_TEMPLATE.md`](./NDA_TEMPLATE.md).
**TODO(avidullu): have counsel review, fill the blanks, and have Roora sign.**
122 - Roora contact on file: `info@rooragrp.com` (HSR Layout, Bangalore).
123
124 ## 9. Pricing request (please quote)
125 1. Price **per rally** and **per minute** of footage.
126 2. **Pilot** price (3 short videos, ~30?36 rallies) as a paid calibration round.
127 3. Indicative **scale** price for ~36?45 longer videos.
128 4. Turnaround for the pilot and throughput (rallies/day) at scale.
129 5. QA process and rework policy.

```

130 6. Whether annotators have racket-sport familiarity (helpful, not required).
131 7. Confirmation you can deliver the CSV schema above.
132

24. user (2026-06-12T09:03:51.749Z)

```
1      # Illustrated Annotation Guide ? Rally Labeling
2
3      Supplement to [R00RA_BRIEF.md](./R00RA_BRIEF.md). Sports: **Badminton, Pickleball,
Table Tennis** (singles **and** doubles).
4
5      > **Illustrations:** diagrams are drawn in text so they survive copy/paste. Boxes marked
6      > `[SCREENSHOT: ...]` are placeholders.
7      > **TODO(avidullu):** replace every `[SCREENSHOT: ...]` placeholder with a real frame grab
8      > from the pilot videos before sending this guide to Roora.**
9
10     ## Part 1 ? What you are producing
11     One CSV per video (from the template Sheet), one row per rally:
12
13     ```
14     rally_number, start_mmss, end_mmss, start_time, end_time, shots_count, ending_reason,
last_shot_outcome, score_before, score_after, notes
15     ```
16
17     You type: `rally_number`, `start_mmss`, `end_mmss`, `shots_count`, `ending_reason`,
18     `last_shot_outcome` (+ optional score/notes). The `start_time`/`end_time` decimal-second
columns
19     **fill themselves** via a formula already in the template.
20
21     For every rally you record:
22     - **When it started** (serve contact) ? `start_mmss` (`m:ss.mmm`, e.g. `1:12.40`)
23     - **When it ended** (shuttle/ball dead) ? `end_mmss`
24     - **How many shots** happened ? `shots_count`
25     - **Why it ended** ? `ending_reason` . **Where the last shot landed** ?
`last_shot_outcome`
26
27     Everything else (gaps between rallies) you **ignore**.
28
29     ## Part 2 ? The three golden rules
30
31     ### Rule 1: Ignore all dead time
32     A rally is **continuous active play**. The moment play stops, the rally is over. The
time
33     until the next serve is **dead time** and is never labeled.
34
35     ```
36     DEAD          RALLY 1          DEAD          RALLY 2          DEAD
37     (ignore) |=====| (ignore) |=====| (ignore)
38             ^               ^             ^               ^
39             serve          shuttle        serve          shuttle
40             contact        lands          contact        lands
41             =start_1       =end_1         =start_2       =end_2
42     ```
43
44     Dead time includes: walking back to serve, picking up the shuttle/ball, wiping sweat,
45     bouncing the ball before serving, the score being announced, replays/slow-mo, crowd or
coach
46     shots, changeovers, towel/medical timeouts, and **pre-match / between-games warm-up
47     (knock-up)** ? see the Rule 2 caveat below.
48
49     ### Rule 2: Include every rally ? even the ugly ones
50     Every served point gets a row, including:
51     - **Service faults** (serve into net/out/illegal) ? usually `shots_count = 1`,
`ending_reason = service_fault`.
52     - **Very short rallies** (serve + one error).
```

```

53 - **Lets / replays** ? `ending_reason = let`; the replayed point is its **own** next
row.
54
55 A **missed rally is the worst error** in this project ? the model must learn that short
and
56 failed rallies are still rallies.
57
58 > **?? BUT warm-up / knock-up is NOT a rally** ? a common trap, because Rule 2 says
"include
59 > everything." Pre-match and between-games warm-up *looks* like rallying but is not
scored play;
60 > **do not label it.** How to tell it apart:
61 > - **No real serve.** Real play starts with a proper serve from the service position
(diagonal,
62 > behind the service line). Warm-up has players feeding cooperatively from mid-court
with no
63 > serve ? not a rally.
64 > - **Cooperative, not competitive.** Gentle hits straight to each other; nobody is
trying to win
65 > the point (no smashes-to-win, no scrambling/diving).
66 > - **Score isn't progressing.** No umpire / scoreboard not running / players casually
retrieve
67 > and re-feed; the score doesn't change. Often **several shuttles/balls** in use at
once.
68 > - **Per sport:** badminton & pickleball ? real play starts with the underhand serve
from the
69 > service court / behind the baseline; table tennis ? real play starts with the
tossed,
70 > two-bounce serve (knock-up is cooperative rallying without it).
71 >
72 > **When unsure:** find the first proper serve where the score starts to progress ?
labeling
73 > begins there. If you genuinely can't tell whether the match has started, **flag the
clip to us
74 > ? don't guess.**
75
76 ### Rule 3: A shot = one contact with the racket / paddle / bat
77 Count every time the shuttle/ball touches a racket, paddle, or bat. The serve is shot 1.
78 - **Count:** any racket/paddle/bat contact (serve, return, smash, block?).
79 - **Don't count:** floor bounces, table bounces, net clips, the toss before a serve.
80
81 5-shot example: `serve(1) ? return(2) ? hit(3) ? hit(4) ? winner(5)` ? `shots_count =
5`, `ending_reason = winner`.
82
83 ## Part 3 ? Where exactly does a rally start and end?
84 **START (`start_mmss`):** the exact frame the server's racket/paddle/bat **contacts**
the
85 shuttle/ball. Not the toss, not the wind-up ? the contact frame.
86 `[SCREENSHOT: serve-contact frame]`
87
88 **END (`end_mmss`):** the exact frame the shuttle/ball becomes **dead**:
89 - Badminton: shuttle touches the floor, or play stops on a fault.
90 - Pickleball: the second floor bounce, or the frame the ball hits the net / lands out.
91 - Table tennis: the second bounce on one side, a non-serve net hit, or off the table.
92
93 `[SCREENSHOT: end frame]`
94
95 If occluded/off-screen, pick the closest clear frame and add a note.
96 **Boundary tolerance: ±0.3 s** (~±9 frames at 30 fps).
97
98 ## Part 4 ? ending_reason: how to choose
99 Pick exactly **one** value for how the rally ended. Stop at the first match:
100
101 1. Did the **serve** itself fail / was illegal (and ended the point)? ? `service_fault`
102 2. Was the point **replayed** (let / interruption)? ? `let`

```

```

103     3. Did the final shot land **in** and the opponent make **no legal contact**? ? `winner`
(a clean winner)
104     4. Did the losing player make an error (netted / out / missed) **under pressure** ?
105         stretched, wrong-footed, or rushed by a strong shot? ? `forced_error`
106     5. Did the losing player make an error on a **routine, unpressured** ball they had
107         the time and position to make? ? `unforced_error`
108     6. None of the above / cannot tell ? `other` (add a note)
109
110     **Forced vs unforced ? the one judgment call.** Decide using the opponent's *preceding*
111     shot:
112     - `forced_error` = the opponent's shot **caused** the miss (pace, placement, deception;
113         the player barely reached it or was pulled out of position).
114     - `unforced_error` = a clean, makeable ball the player missed on their own.
115     - When genuinely unsure between the two, choose `unforced_error` and add a brief note.
116
117     Notes:
118     - A clean `winner` means **no legal contact** by the opponent. If they got a racket /
paddle /
119         bat on it but netted or hit out, that is a `forced_error` / `unforced_error` (their
shot), not a `winner`.
120     - A serve into the net is `service_fault` (rule 1 beats everything).
121
122     ### Then set `last_shot_outcome` ? where the last shot physically went
123     This is a **separate, observational** column (no judgment): just look at the last shot.
124     - `in` ? landed in (winners; also a put-away the opponent couldn't reach).
125     - `net` ? went into the net.
126     - `out` ? landed outside the court / past the lines.
127     - `n_a` ? not applicable: use for `service_fault` and `let`.
128
129     So each rally gets both: **why** (`ending_reason`) and **where** (`last_shot_outcome`) ?
e.g. a
130     smash winner = `winner, in`; a clear forced long = `forced_error, out`; a routine ball
netted =
131     `unforced_error, net`.
132
133     ## Part 5 ? Per-sport detail
134
135     ### Badminton
136     - **Serve:** underhand, below the waist, diagonal. Start = racket?shuttle contact.
137     - **Rally end:** shuttle hits the floor, or play stops on a fault.
138     - **Shots:** every racket?shuttle contact; serve = shot 1.
139     - **Service faults** (1-shot rally, `service_fault`): serve into net, serve outside
service court, contact above waist, foot fault.
140
141     Worked example:
142     ```
143     [SCREENSHOT: badminton serve]
144     Rally 4: serve 1:28.30, long rally, smash winner, shuttle lands 1:44.10, 12 contacts.
145         4, 1:28.30, 1:44.10, (auto), (auto), 12, winner, 3-5, 4-5,
146     Rally 5: serve 1:57.00 straight into the net.
147         5, 1:57.00, 1:57.90, (auto), (auto), 1, service_fault, 4-5, 4-6,
148     ```
149
150     **Badminton ? DOUBLES (2 players per side).** All rally rules are **identical** to
singles:
151     - `start` = serve contact; `end` = shuttle down / fault; a **shot = any racket?shuttle
contact by either player on a side**.
152     - `shots_count` = **total** contacts across both teams in the rally. **Do not track
which player hit** ? just count every contact.
153     - Net exchanges (drives, flat pushes) come fast ? **frame-step** so you neither miss nor
double-count contacts.
154     - Only the **designated receiver** returns the serve; if the wrong player returns, the
rally ends on a fault ? still a row (`service_fault` if it's a serve/receiving-position fault,
else `other` + note).
155     - Two teammates may **not** both hit the shuttle in succession (double-hit). If you see

```



```

two contacts on the same side back-to-back, that's a fault ending the rally ? count both
contacts, then the rally ends.
156
157     ### Pickleball
158     - Serve: underhand, paddle below waist, hit diagonally; must clear the non-volley
zone (the "kitchen"). Start = paddle?ball contact.
159     - Two-bounce rule: the serve must bounce once in the receiver's court and the return
must bounce too, before either side may volley. These bounces are normal play, not rally
ends, and bounces are not shots.
160     - Rally end: ball bounces twice, lands out, hits the net (fails to cross), or a
fault (e.g. volleying in the kitchen).
161     - Shots: every paddle?ball contact only; serve = shot 1; floor bounces are not
shots.
162     - Service faults (`service_fault`): serve into net, out/long, short in the kitchen,
foot fault.
163
164     Worked example:
165     ```
166     [SCREENSHOT: pickleball serve + kitchen line]
167     Rally 2: serve 0:33.10; serve bounces, return bounces, a few volleys; receiver nets an
easy one at 0:41.85; 6 paddle contacts.
168     2, 0:33.10, 0:41.85, (auto), (auto), 6, unforced_error, 1-0, 2-0, easy ball netted
169     ```
170
171     Pickleball ? DOUBLES (the standard format, 2 per side). All rally rules
identical to singles:
172     - `start` = serve contact; a shot = any paddle?ball contact by either teammate;
floor bounces still not shots; two-bounce rule still applies.
173     - `shots_count` = total paddle contacts across both teams; don't track which player.
174     - Serving sequence (FYI only ? you do NOT annotate it): both partners serve before
the serve passes to the opponents, except the very first service turn of the game. The called
score has three numbers (server score ? receiver score ? server #1/#2). Ignore this for
annotation; just mark boundaries / shots / ending.
175     - If both teammates contact the ball in succession, that's a fault ? rally ends; count
the contacts, set the ending per the fault.
176
177     ### Table tennis
178     - Serve: ball tossed up from an open palm, then struck so it bounces once on each
side. Start = the racket?ball strike (not the toss).
179     - Rally end: a player fails a legal return ? ball double-bounces on their side, goes
off the table, or is hit into the net on a non-serve shot.
180     - Shots: every racket?ball contact only; serve = shot 1; table bounces are not
shots.
181     - Let: a serve that clips the net but is otherwise legal ? the point is replayed.
`ending_reason = let`, and the replay is its own next row.
182     - Service faults (`service_fault`): missed/illegal toss, serve into net, serve off
the table.
183     - Speed warning: contacts can be < 0.2 s apart ? step frame-by-frame to count
shots.
184     - Doubles (if a clip appears): partners must hit in strict alternation and serve
diagonally; a missed alternation is a fault that ends the rally. Same rule ? count every
racket?ball contact as a shot; `shots_count` is the team total.
185
186     Worked example:
187     ```
188     [SCREENSHOT: table-tennis serve contact]
189     Rally 9: serve 4:10.40; fast 7-contact rally; forehand winner, opponent no contact; dead
4:14.05.
190     9, 4:10.40, 4:14.05, (auto), (auto), 7, winner, 6-9, 6-10,
191     Rally 10: serve 4:22.10 clips the net (otherwise good) -> let.
192     10, 4:22.10, 4:23.00, (auto), (auto), 1, let, 6-10, 6-10, net-cord serve, replayed
193     ```
194
195     ## Part 6 ? Edge cases & FAQ
196     - Two rallies with almost no gap? Still two rows. `end_time[N] <= start_time[N+1]`,

```

never overlap.

```
197     - **Interrupted & replayed?** Label the interrupted attempt as its own row
(`ending_reason=let`), then the replayed point as the next row.
198     - **Serve tossed but caught / re-served (no contact)?** No contact = no rally yet. Start
at the actual serve contact; don't label the aborted toss.
199     - **Opponent reached the ball but netted/hit out?** That's a
`forced_error`/`unforced_error` (their shot), not `winner` ? `forced` if your shot pressured
them, `unforced` if it was routine.
200     - **Can't read the scoreboard?** Leave `score_before`/`score_after` blank. Never guess.
```

25. assistant (2026-06-12T09:03:54.588Z)

Now let me search more specifically for any evaluation or comparison data with Roora annotations, and look for scratch notes or reports:

26. user (2026-06-12T09:05:55.895Z)

Command running in background with ID: bixcw3i75. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\tasks\bixcw3i75.output. You will be notified when it completes. To check interim output, use Read on that file path.

27. assistant (2026-06-12T09:05:58.404Z)

Let me check the memory and check if there are any recent notes or reports:

28. user (2026-06-12T09:05:58.600Z)

EISDIR: illegal operation on a directory, read 'C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory'

29. user (2026-06-12T09:06:01.917Z)

MEMORY.md
audio-e1-findings.md
candidate-segments-finetuning.md
code-map-artifact.md
collector-indexer-bridge.md
current-state.md
feedback-tail-long-job-logs.md
golden-set-vendoring.md
monetization-plan.md
monetization-research-status.md
session-handoff.md
sibling-internship-repo.md
tracknet-wasb-wsl2-decision.md
working-agreement-attribution.md
working-agreement-checkpoints.md

30. user (2026-06-12T09:06:11.453Z)

```
1      ---
2      name: golden-set-vendoring
3      description: Plan to generate golden-set rally labels via third-party vendors + an
extensible regression-testing flow
4      metadata:
5        node_type: memory
6        type: project
7        originSessionId: aa329f09-28b7-4acc-b062-98ec7e905abc
8      ---
9
10     How we produce trusted **golden-set ground truth** (hand-verified rally `[start,end)`
labels) and use it to regression-test the production tool. Full design + vendor shortlist is
version-controlled in `docs/GOLDEN_SET_VENDORING.md` (and ADR-006); this file holds the durable
```

decisions.

11

12 ****Keystone insight:**** a vendor's label CSV is structurally identical to a pipeline prediction, so the existing ``backend/eval/metrics.py`` (``interval_iou`?```match_intervals`?```score``) grades all three of: product-vs-GT, vendor-vs-answer-key, and annotator-vs-annotator (IAA). No new scoring code ? only a new *source* of intervals.

13

14 ****Decisions locked with the user (2026-05-30):****

15 - ****Licensed/owned video only**** ? vendors never see user uploads (this is the monetization ****privacy lever****, see `[[monetization-plan]]`). Start corpus = the user's own growing match collection + ShuttleSet.

16 - ****Rally `[start,end)` depth**** (cheapest; extensible later to shot-counts/(x,y)).

17 - ****Tiered turnaround:**** slow human tier (authoritative) + fast automated tier (CI / "surprise regression"). ?? ****Oracle rule:**** the fast-tier auto-labeler MUST be a different model lineage than production, else shared blind spots ? false green.

18 - ****Vendor bake-off****, favouring ****Bangalore/India-local**** (user is Bangalore-based). ****Superseded 2026-06-02 ? started as a single-vendor **Roora** pilot; see Status below.****

19

20 ****Architecture (partly built ? 2026-06-02):**** the ``AnnotationProvider`` registry + ``FileProvider`` (GS-1), the ``regress(video)`` runner (GS-2 ? ``backend/eval/regression.py`` + ``run_regression.py``), and the ``AutomatedProvider`` oracle scaffold (GS-4) have ****landed****. The vendor-dependent ****`HumanVendorProvider`** (GS-3)****** is the remaining slice, parked until the Roora pilot confirms quality.

21

22 ****Vendor leads (verified India presence + video; pricing NOT verified ? all quote-only):**** primary = ****iMerit**** (Bengaluru office, sports domain, enterprise QA) and ****Taskmonk**** (Bengaluru, platform+service, likely cheaper/agile); heavyweight third = ****TELUS/Playment**** (Bangalore-founded CV). Excluded: ****Karya**** (Bengaluru but speech/text only, no video). Tier-2 leads + confidence labels in the doc. (PR #12.)

23

24 ****Status (2026-06-02):****

25 - ****Vendor selected for the initial pilot: Roora**** (Roora ML Pvt Ltd, HSR Layout, Bengaluru; ``info@rooragr.com``) ? site/capability (image/video/text annotation; sports industry) and location verified; ****quality TBD via the pilot****. Run as a ****single-vendor pilot first****, NOT a multi-vendor bake-off; iMerit/Taskmonk/TELUS are the fallback if it misses the quality bar.

26 - ****Annotation guideline authored**** (Step 0): collector repo ``docs/annotation/`` ? ``ROORA_BRIEF.md`` (rally CSV schema: ``rally_number,start_time,end_time,shots_count,ending_reason`, mm:ss?seconds``), ``ANNOTATION_GUIDE.md`` (illustrated; three golden rules = ignore dead time / include EVERY rally incl. service faults & lets / shot = one racket-paddle-bat contact; ``ending_reason`` = forced/unforced-error taxonomy (``winner | forced_error | unforced_error | service_fault | let | other``, shared across net-separated sports; superseded observational ``into_net`/`out`` 2026-06-06); ****singles & doubles**** for badminton & pickleball), + intake-manifest & NDA templates.

27 - ****Pilot scope = 3 sports**** (badminton, pickleball, table tennis), scaling to ~12?15 videos/sport. The ****collector now ingests all three**** via ``import-local`` (collector ADR-001 keeps explicit per-sport code over a registry; collector test coverage 87%).

28 - ****Still pending:**** the internal ****gold key**** (5?10 expert-labeled clips) and ****GS-3 `HumanVendorProvider`****.

29

31. assistant (2026-06-12T09:06:14.849Z)

Let me check if the background command completed and search the code for any timing offset/delay parameters:

32. user (2026-06-12T09:06:16.267Z)

C:\Users\avidu\Projects\.NET Aspire\aspire/.agents/skills/pr-testing/SKILL.md
C:\Users\avidu\Projects\.NET Aspire\aspire/docs/contributing.md
C:\Users\avidu\Projects\Annotation Setup\golden_labels_badminton.csv
C:\Users\avidu\Projects\Annotation Setup\Roora Pilot Review ? Badminton (2026-06-06).md
C:\Users\avidu\Projects\Annotation Setup\Roora Pilot Review ? Pickleball & Table Tennis (2026-06-10).md

C:\Users\avidu\Projects\badminton-highlight-indexer/docs/archives/decisions/DECISIONS.md
C:\Users\avidu\Projects\badminton-highlight-indexer/docs/archives/quality-iterations/2026-06-multivideo-lovo/README.md
C:\Users\avidu\Projects\badminton-highlight-indexer/docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md
C:\Users\avidu\Projects\badminton-highlight-indexer/docs/BACKLOG.md
C:\Users\avidu\Projects\badminton-highlight-indexer/docs/CODE_MAP.md
C:\Users\avidu\Projects\badminton-highlight-indexer/docs/COMMERCIALIZATION.md
C:\Users\avidu\Projects\badminton-highlight-indexer/docs/EVALUATION.md
C:\Users\avidu\Projects\badminton-highlight-indexer/docs/GOLDEN_SET_IMPLEMENTATION.md
C:\Users\avidu\Projects\badminton-highlight-indexer/docs/GOLDEN_SET_PHASE1_VIDEOS.md
C:\Users\avidu\Projects\badminton-highlight-indexer/docs/GOLDEN_SET_VENDORING.md
C:\Users\avidu\Projects\badminton-highlight-indexer/docs/HOW_RALLY_DETECTION_WORKS.md
C:\Users\avidu\Projects\badminton-highlight-indexer/docs/NEXT_STEPS.md
C:\Users\avidu\Projects\badminton-highlight-indexer/docs/ROADMAP.md
C:\Users\avidu\Projects\badminton-highlight-indexer/README.md
C:\Users\avidu\Projects\badminton-highlight-indexer/scratch/HIGH_RISK_REMEDIATION_PLAN.md

33. assistant (2026-06-12T09:06:18.148Z)

Perfect! I found the key files: the "Roora Pilot Review" documents. Let me read those:

34. user (2026-06-12T09:06:18.609Z)

```
1      # Roora Pilot ? Badminton Review & Conventions for Scale-up
2
3      **Date:** 6 June 2026
4      **Re:** Phase-A calibration pilot ? badminton job (CVAT job id 4075945)
5      **Companion docs:** *Roora Annotation Brief ? Rally Golden Set (v2)*, *Illustrated
Annotation Guide*
6
7      ---
8
9      ## 1. Summary
10
11     Thank you ? the first badminton delivery is a solid calibration round. We've built an
ingestion
12     adapter on our side, so **your CVAT export works for us directly** and the timing/format
mechanics
13     are handled here, not in your workflow. After a full frame-by-frame review, the rally
**timings
14     are generally good** ; the items that genuinely need your attention are about
**annotation
15     completeness and rally boundaries** ? and they matter most for the larger batches coming
next.
16
17     ---
18
19     ## 2. Handled on our side ? no change needed from you
20
21     - **Export format.** Keep working in **CVAT** ; we ingest your bounding-box export as-is.
No need
22     to produce a CSV.
23     - **Timestamps.** We recompute each rally's time from the box **frame range at the true
fps**
24     (`seconds = frame / fps`), so the CVAT time attributes don't matter, and `mm:ss` vs
decimal is
25     a non-issue ? our ingestion parses both. Minor boundary slack is absorbed by padding
downstream.
26     - **Per-frame label drift.** Where a rally's boxes carried slightly different attribute
values
27     across frames, we collapse to one value per rally automatically.
28
29     ## 3. What needs your attention (for the upcoming larger batches)
30
```

```

31     These can't be fixed in code ? they're annotation-quality conventions.
32
33     ### 3.1 Completeness ? don't miss rallies *(most important)*
34     Our review found rallies with **no label**:
35     - **~1:46** ? a 5-shot rally omitted entirely (frame grab attached).
36     - **3:34?3:43** ? a full rally omitted because it was **absorbed into the previous
rally's box**
37     (the box ran from 3:25 all the way to 3:43, covering two rallies + the dead time
between).
38
39     The brief calls a missed rally "the most serious error." For 30-min videos this is the
#1 thing
40     to get right.
41
42     ### 3.2 One rally = one continuous interval
43     A few boxes spanned **past the shuttle-dead frame** or **across two rallies**:
44
45     | rally | delivered span | true rally | issue |
46     |-----|-----|-----|-----|
47     | 2 | 0:02?0:32 | 0:29?0:32 | box started ~27 s into dead time |
48     | 7 | 1:35?1:51 | 1:35?1:38 | box ran ~13 s past the end |
49     | 16 | 3:25?3:43 | 3:25?3:29 | box covered a **second** rally (see 3.1) |
50
51     Convention: start the box at the **serve-contact** frame, end it at the **shuttle-dead**
frame,
52     and start a **new** box for the next rally ? never let one box bridge a gap.
53
54     ### 3.3 Endings ? `winner` vs an error
55     A `winner` means the opponent made **no legal contact**. A few rallies where the shuttle
landed
56     **out** were marked `winner` (e.g. 6, 7, 9). If the rally ended on a miss, it's a
57     `forced_error`/`unforced_error`, not a winner.
58
59     ### 3.4 New field ? `last_shot_outcome` *(small label-schema add)*
60     We're adding one attribute so the physical outcome is captured alongside the reason:
61
62     | field | values |
63     |-----|-----|
64     | `ending_reason` | winner \| forced_error \| unforced_error \| service_fault \| let \|
other \|
65     | `last_shot_outcome` | in \| net \| out \| n_a |
66
67     Just add `last_shot_outcome` to the CVAT label schema (one dropdown). If that's awkward,
tell us
68     and we'll keep it optional.
69
70     ### 3.5 Scope ? pickleball & table tennis
71     The pilot is 1 badminton + 1 pickleball + 1 table tennis; we've received badminton only.
Please
72     confirm status / ETA for the other two.
73
74     ---
75
76     ## 4. Convention checklist (for the next batches)
77
78     - [ ] **Every** served point labeled ? no missed rallies, including back-to-back ones.
79     - [ ] One box per rally: serve-contact ? shuttle-dead; never bridge a gap or two
rallies.
80     - [ ] `ending_reason` ? {winner, forced_error, unforced_error, service_fault, let,
other}.
81     - [ ] `last_shot_outcome` ? {in, net, out, n_a} (one dropdown).
82     - [ ] `shots_count` = total racket contacts (serve = 1).
83     - [ ] CVAT export is fine ? we handle the rest.
84
85     *(Open items on our side to confirm with you: which brief version your team worked from,

```

and that
86 the per-video fps will be in the manifest going forward.)*
87

35. user (2026-06-12T09:06:18.692Z)

```
1      # Roora Pilot Review ? Pickleball & Table Tennis
2
3      **Date:** 10 June 2026
4      **Re:** Phase-A calibration pilot ? pickleball + table tennis deliveries (CVAT exports;
5      `Video 2 - Pickleball.mp4`, `Video 3 - Tabletennis.mp4`)
6      **Companion docs:** *Roora Annotation Brief ? Rally Golden Set (v2)*, *Illustrated
7      Annotation Guide (v2)*, *Pilot Review ? Badminton (2026-06-06)*
8
9      ---
10
11     ## 1. Summary
12
13     Thank you for completing the pilot trio ? both deliveries ingest cleanly, shot counts
14     are filled and plausible, and numbering and rally separation are clean. However, this round has
15     **one serious issue: rally boundary accuracy.** Every rally's start/end in both videos is off by
16     **0.5 s or more ? up to ~2 s in several** ? against the agreed tolerance of ** $\pm 0.3$  s** (Brief
17     §4, Guide Part 3). The badminton round's timings were generally good, so this is a regression;
18     we'd like to understand what changed in the process. The sections below explain the impact, the
19     root cause we found in the files, and the conventions to lock before the Phase-B batches.
20
21     ## 2. Boundary accuracy ? outside the agreed tolerance *(most important)*
22
23     **What we found.** A full frame-by-frame re-check of all 12 rallies shows every
24     `start`/`end` boundary off by 0.5 s or more, several by up to ~2 s.
25
26     **Root cause (from your files).** All 24 rally boundaries sit **exactly on the 30-frame
27     sampling grid** ? at 59.94 fps that grid is ~0.5 s apart, so a boundary placed only on grid
28     frames *cannot* meet  $\pm 0.3$  s. Sampling every 30th frame **inside** a rally is fine; but the
29     **first box must be on the exact serve-contact frame and the last box on the exact ball-dead
30     frame** (Guide Part 3), even when those fall between grid steps.
31
32     **Why this matters so much.** These labels are the golden ground truth we train and
33     evaluate a rally-segmentation model against ? **boundary precision is the product itself**. The
34     rallies in these videos run 1?8.5 s, so a 0.5?2 s boundary error corrupts 10?100% of a rally's
35     span; the 1.0 s table-tennis rally #2 can be erased entirely by it. Concretely: at our standard
36     evaluation gate (overlap ? 50%), a typical 4.5 s rally labeled 2 s late **fails to match even a
37     perfectly correct detection** ? the dataset would score correct model output as wrong and wrong
38     output as right. Imprecise gold is worse than no data, because it silently corrupts every model
39     trained on it and every quality number reported from it.
40
41     **Going forward:** place the first/last box of every rally on the exact contact/dead
42     frame (frame-step to find it ? Guide Parts 3 and 6), and self-check boundaries against  $\pm 0.3$  s
43     before submitting (Guide Part 7 checklist).
44
45     ## 3. Rallies already in progress when the video starts ? skip them
46
47     Pickleball **rally #1 starts at the very first frame of the file (0:00.000)**. The serve
48     contact is not in the video ? the file begins mid-rally ? so its start time is unknowable and
49     the rally should not have been labeled at all.
50
51     **Convention to follow from now on:** if a video begins during an ongoing rally, **skip
52     that rally entirely and start annotating from the first rally that starts (serve contact
53     visible) within the video.** Never extrapolate or guess a boundary you did not see. The same
54     applies at the end of a file: if the video cuts off mid-rally, skip that rally. When in doubt,
55     flag it to us rather than guessing (Guide Part 6).
56
57     ## 4. Reminder for Phase B ? dead time and warm-up
58
59     Phase-B videos are ~30 minutes and will contain substantial non-match footage, so we
```

want to re-emphasize two rules now (Guide Rule 1 + Part 6):

```
32
33     - **Skip all dead time** ? before the match, between rallies, during breaks: walking,
picking up the ball, serve preparation, changeovers, coaching. None of it is ever labeled.
34     - **Skip warm-up / knock-up rallies and practice points** ? they are not real rallies.
Labeling starts at the **first real, scored serve**. When in doubt whether the match has
started, flag it to us.
35
36     ## 5. Schema hygiene ? remove the fields we did not ask for
37
38     Please remove the following from the CVAT label schema; they add noise and, in one case,
actively confusing data:
39
40     - **`start time` / `end time`** ? the delivered values are decimal minutes computed at
an assumed 15 fps, ~4x off from real time (the videos are 59.94 fps). We derive all timing from
the box frame ranges, so these fields serve no purpose and can only mislead. Remove them.
41     - **`score_before(A-B)` / `score_after(A-B)`** ? empty on every box in both videos. We
are dropping scores from the schema; remove them.
42
43     The schema going forward is exactly: **`rally_number`, `shot_count`, `ending_reason`,
`last_shot_outcome`, `note`** ? nothing else.
44
45     ## 6. Field quality
46
47     - **`last_shot_outcome`** ? pickleball has the literal value `0` on all 6 rallies (not a
valid value); table tennis left it blank. The vocabulary is `in | net | out | n_a` ? please use
it.
48     - **`ending_reason` = `other` is overused** ? 5 of 6 table-tennis rallies. `other` is
the *last resort* (Guide Part 4): "out of table" and "net" endings map to
`forced_error`/`unforced_error` (with `last_shot_outcome` = `out`/`net`). Also, a note is
**required** whenever `other` is used ? 3 of the 5 `other` rallies have none.
49     - **One value per rally** ? table-tennis rally #5 carries `other` on some frames and
`winner` on others. Attribute values must be identical on every box of a rally.
50
51     ## 7. What was good ? please keep doing it
52
53     - In-rally sampling discipline is excellent: boxes every 30 frames with zero gaps in all
12 rallies.
54     - Clean sequential numbering, no overlapping rallies.
55     - `shot_count` filled everywhere and plausible against rally durations (including the
fast 12-shot table-tennis rally).
56     - Both sports delivered together, in a format our ingestion reads directly.
57
58     ## 8. Before Phase B
59
60     - We will fold the conventions above (exact first/last frames, skip rallies in progress
at file start/end, dead-time and warm-up rules, the reduced field schema) into the next revision
of the Guide ? this review is the joint calibration loop the Brief (§3, Phase A) describes.
61     - Please confirm which version of the Brief/Guide your team worked from on these two
videos.
62
```

36. assistant (2026-06-12T09:06:21.339Z)

Excellent! I found the documentation. Now let me search for any related eval scripts or metrics files that might discuss the offset:

37. user (2026-06-12T09:06:22.259Z)

```
backend\eval\calibrate_local.py:77:def run(specs: List[Dict], iou: float = 0.5, metric: str =
"f1") -> dict:
backend\eval\calibrate_local.py:117:     ap.add_argument("--iou", type=float, default=0.5)
backend\eval\calibration.py:41:def evaluate(preds: List[Interval], gts: List[Interval],
iou_threshold: float = 0.5) -> Score:
backend\eval\calibration.py:62:         metric: str = "f1", iou_threshold: float = 0.5)
```

```

-> Tuple[Config, float]:
backend\eval\calibration.py:73:         k: int = 5, metric: str = "recall",
iou_threshold: float = 0.5) -> Dict[str, Any]:
backend\eval\calibration.py:114:         metric: str = "f1", iou_threshold: float
= 0.5) -> Dict[str, Any]:
backend\eval\calibration.py:149:         gts: List[Interval], iou_threshold: float
= 0.5) -> Dict[str, Any]:
backend\eval\classifier.py:81:     def predict(self, X, threshold: float = 0.5) -> np.ndarray:
backend\eval\calibrate_wasb.py:67:         iou: float = 0.5) -> dict:
backend\eval\calibrate_wasb.py:101:     ap.add_argument("--iou", type=float, default=0.5)
backend\eval\distill_local.py:43: PROPOSAL = (0.005, 1.0, 0.5)
backend\eval\distill_local.py:80: def load_video(spec: Dict, iou: float = 0.5) -> Dict:
backend\eval\distill_local.py:90:         threshold: float = 0.5) -> List[Interval]:
backend\eval\distill_local.py:98: def run(specs: List[Dict], iou: float = 0.5, threshold: float =
0.5,
backend\eval\distill_local.py:150:     ap.add_argument("--iou", type=float, default=0.5)
backend\eval\distill_local.py:151:     ap.add_argument("--threshold", type=float, default=0.5,
help="classifier probability cutoff")
backend\eval\audio\e1_probe.py:54:         k: float = 2.5, cluster_gap: float = 2.0, min_hits:
int = 2, iou: float = 0.5) -> dict:
backend\eval\audio\e1_probe.py:108:     ap.add_argument("--iou", type=float, default=0.5)
backend\eval\export_wasb_segments.py:36: # F1 0.588 (baseline) -> 0.746. Override on the CLI to
export a different operating point.
backend\eval\export_wasb_segments.py:85:         iou_threshold: float = 0.5) ->
dict:
backend\eval\export_wasb_segments.py:87:     plus the boundary errors and any extra (false-
positive) tuned segments."""
backend\eval\export_wasb_segments.py:141:         iou: float = 0.5, baseline: bool = False,
backend\eval\export_wasb_segments.py:184:         print(f"mean boundary error: start
{sc.mean_start_error:.2f}s | end {sc.mean_end_error:.2f}s")
backend\eval\export_wasb_segments.py:207:     ap.add_argument("--iou", type=float, default=0.5)
backend\eval\eval_partial_labels.py:70:         gate_crossings: int = 2, iou: float = 0.5,
backend\eval\eval_partial_labels.py:138:         for mwd in (0.5, 1.0, 1.5, 2.0)
backend\eval\eval_partial_labels.py:144:         gate_options: Tuple[int, ...] = (0, 2), iou:
float = 0.5,
backend\eval\eval_partial_labels.py:201:     ap.add_argument("--iou", type=float, default=0.5)
backend\eval\harness.py:67:         stages: List[str], iou_threshold: float = 0.5,
backend\eval\harness.py:116:         f"{'Prec':>6} {'Rec':>6} {'F1':>6} {'mIoU':>6} {'dStart':>7}
{'dEnd':>7}",
backend\eval\metrics.py:49:         iou_threshold: float = 0.5) -> MatchResult:
backend\eval\metrics.py:105:         iou_threshold: float = 0.5,
backend\eval\metrics.py:107:         """Compute precision/recall/F1, mean matched IoU, and boundary
errors.
backend\eval\rally_seq_proto.py:19:     -> merge tiny gaps / drop short -> score vs GT at
IoU>=0.5 (same metric as the heuristic).
backend\eval\rally_seq_proto.py:22: hand features). It exists to answer one question: is the
rally boundary learnable from
backend\eval\rally_seq_proto.py:68:     instead of 0.5, and dead-time frames produce identical
all-zero feature rows (guaranteed
backend\eval\rally_seq_proto.py:70:     pos, neg = p[y > 0.5], p[y <= 0.5]
backend\eval\rally_seq_proto.py:74:     return float((ranks[y > 0.5].sum() - pos.size * (pos.size
+ 1) / 2) / (pos.size * neg.size))
backend\eval\rally_seq_proto.py:105:         f1s.append(score(segs, v["gts"], 0.5).f1)
backend\eval\rally_seq_proto.py:134:         sc = score(segs, v["gts"], 0.5)
backend\eval\rally_seq_proto.py:138:         key=lambda t:
score(probs_to_segments(probs, v["times"], float(t)), v["gts"], 0.5).f1)
backend\eval\rally_seq_proto.py:139:         osc = score(probs_to_segments(probs, v["times"],
float(othr)), v["gts"], 0.5)
backend\eval\regression.py:105:         stage: str = "final", iou_threshold: float = 0.5,
backend\eval\regression.py:162:         iou_threshold: float = 0.5,
backend\eval\training.py:51:         iou_threshold: float = 0.5) -> List[int]:
backend\eval\training.py:67:     iou_threshold: float = 0.5,
backend\eval\__init__.py:7: - `metrics` ? pure interval-matching math (IoU, matching, P/R/F1,
boundary error).

```


38. assistant (2026-06-12T09:06:24.336Z)

Let me look at the metrics.py file to understand how boundary errors are calculated:

39. user (2026-06-12T09:06:24.734Z)

```
1      """Pure temporal interval-matching metrics for rally evaluation.
2
3      No I/O, no DB, no global state ? every function here is deterministic and unit
4      tested. An "interval" is a `(start, end)` tuple of floats in seconds with
5      start < end. We score a list of *predicted* rally intervals against a list of
6      *ground-truth* intervals via temporal IoU and one-to-one greedy matching.
7      """
8
9      from dataclasses import dataclass, field
10     from typing import List, Tuple, Optional
11
12     Interval = Tuple[float, float]
13
14
15     def interval_iou(a: Interval, b: Interval) -> float:
16         """Temporal intersection-over-union of two intervals.
17
18         Returns 0.0 when they don't overlap (or either is degenerate). IoU is
19         overlap / union, both measured as durations on the timeline.
20         """
21         a_start, a_end = a
22         b_start, b_end = b
23         if a_end <= a_start or b_end <= b_start:
24             return 0.0
25         inter = max(0.0, min(a_end, b_end) - max(a_start, b_start))
26         if inter <= 0.0:
27             return 0.0
28         union = (a_end - a_start) + (b_end - b_start) - inter
29         return inter / union if union > 0 else 0.0
30
31
32     @dataclass
33     class Match:
34         """One matched predicted?ground-truth pair."""
35         pred_index: int
36         gt_index: int
37         iou: float
38
39
40     @dataclass
41     class MatchResult:
42         """Outcome of matching predictions to ground truth at a given IoU threshold."""
43         matches: List[Match] = field(default_factory=list)
44         unmatched_pred_indices: List[int] = field(default_factory=list) # false positives
45         unmatched_gt_indices: List[int] = field(default_factory=list)   # false negatives
46 (misses)
47
48     def match_intervals(preds: List[Interval], gts: List[Interval],
49                       iou_threshold: float = 0.5) -> MatchResult:
50         """Greedy one-to-one matching of predictions to ground truth by descending IoU.
51
52         Every candidate (pred, gt) pair with IoU >= threshold is considered; pairs
53         are claimed highest-IoU first, and each prediction/GT can be used once. This
54         keeps a single prediction from being credited for two ground-truth rallies
55         (and vice versa).
56         """
57         candidates = []
58         for pi, p in enumerate(preds):
```

```

59         for gi, g in enumerate(gts):
60             iou = interval_iou(p, g)
61             # Require real overlap: at iou_threshold=0.0 a plain `iou >= threshold`
would
62             # match completely disjoint intervals (IoU 0.0), fabricating perfect scores.
63             if iou > 0.0 and iou >= iou_threshold:
64                 candidates.append((iou, pi, gi))
65         # Highest IoU first; ties broken by indices for determinism.
66         candidates.sort(key=lambda c: (-c[0], c[1], c[2]))
67
68         used_pred, used_gt = set(), set()
69         matches: List[Match] = []
70         for iou, pi, gi in candidates:
71             if pi in used_pred or gi in used_gt:
72                 continue
73             used_pred.add(pi)
74             used_gt.add(gi)
75             matches.append(Match(pred_index=pi, gt_index=gi, iou=iou))
76
77         unmatched_pred = [i for i in range(len(preds)) if i not in used_pred]
78         unmatched_gt = [i for i in range(len(gts)) if i not in used_gt]
79         return MatchResult(matches=matches,
80                             unmatched_pred_indices=unmatched_pred,
81                             unmatched_gt_indices=unmatched_gt)
82
83
84     @dataclass
85     class Score:
86         """Aggregate metrics for one (predictions vs ground-truth) evaluation."""
87         iou_threshold: float
88         num_pred: int
89         num_gt: int
90         true_positives: int
91         false_positives: int
92         false_negatives: int
93         precision: float
94         recall: float
95         f1: float
96         mean_iou: float          # over matched pairs (0.0 if none)
97         mean_start_error: float  # mean |pred_start - gt_start| over matches, seconds
98         mean_end_error: float    # mean |pred_end - gt_end| over matches, seconds
99
100         def as_dict(self) -> dict:
101             return self.__dict__.copy()
102
103
104     def score(preds: List[Interval], gts: List[Interval],
105              iou_threshold: float = 0.5,
106              match_result: Optional[MatchResult] = None) -> Score:
107         """Compute precision/recall/F1, mean matched IoU, and boundary errors.
108
109         Pass a precomputed `match_result` to avoid re-matching (e.g. when also
110         extracting the failure lists); otherwise matching is done here.
111         """
112         mr = match_result if match_result is not None else match_intervals(preds, gts,
iou_threshold)
113
114         tp = len(mr.matches)
115         fp = len(mr.unmatched_pred_indices)
116         fn = len(mr.unmatched_gt_indices)
117
118         precision = tp / (tp + fp) if (tp + fp) > 0 else 0.0
119         recall = tp / (tp + fn) if (tp + fn) > 0 else 0.0
120         f1 = (2 * precision * recall / (precision + recall)) if (precision + recall) > 0
else 0.0

```

```

121
122         if mr.matches:
123             mean_iou = sum(m.iou for m in mr.matches) / tp
124             mean_start_err = sum(abs(preds[m.pred_index][0] - gts[m.gt_index][0]) for m in
mr.matches) / tp
125             mean_end_err = sum(abs(preds[m.pred_index][1] - gts[m.gt_index][1]) for m in
mr.matches) / tp
126         else:
127             mean_iou = mean_start_err = mean_end_err = 0.0
128
129         return Score(
130             iou_threshold=iou_threshold,
131             num_pred=len(preds), num_gt=len(gts),
132             true_positives=tp, false_positives=fp, false_negatives=fn,
133             precision=precision, recall=recall, f1=f1,
134             mean_iou=mean_iou, mean_start_error=mean_start_err, mean_end_error=mean_end_err,
135         )
136
137
138     def pr_curve(preds: List[Interval], gts: List[Interval],
139                 thresholds: Optional[List[float]] = None) -> List[Score]:
140         """Score across a sweep of IoU thresholds (default 0.1..0.9 step 0.1).
141
142         Lets you see strict-vs-lenient behaviour rather than a single operating point.
143         """
144         if thresholds is None:
145             thresholds = [round(0.1 * i, 1) for i in range(1, 10)]
146         return [score(preds, gts, iou_threshold=t) for t in thresholds]
147

```

40. assistant (2026-06-12T09:06:26.735Z)

Perfect! Now let me search for any other related documentation or eval reports that might have concrete numbers:

41. user (2026-06-12T09:06:27.183Z)

```

C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled\Adarsh and Avi.rallies.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled\gBaaddy20May2026.rallies.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Golden
Labelled\KushagraYashAviFirstVideo.rallies.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Golden
Labelled\KushagraYashAviFirstVideo.report.md
C:\Users\avidu\Projects\Annotation Setup\Collect\Golden
Labelled\KushagraYashAviFirstVideo.summary.md
C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled\LargeTestVideo.rallies.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled\MahadevpuraSingles.rallies.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled\MahadevpuraSingles.report.md
C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled\MahadevpuraSingles.summary.md
C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled\TestLargeVideo.rallies.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\adarsh_avi_full_traj.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\adarsh_avi_traj.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\adarsh_traj.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\gbaaddy_traj.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\kushagra_traj.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\largetestvideo_traj.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\mahadevpura_traj.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\pickleball_traj.csv
C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\roora\annotations.report.md
C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\roora\annotations.summary.md

```

42. assistant (2026-06-12T09:06:30.046Z)

Let me check the Roora report and summary files:

43. user (2026-06-12T09:06:30.264Z)

```
1      # Sports Data Collector ? Delivery / Ingestion Report
2
3      ## Run
4
5      | Field | Value |
6      | --- | --- |
7      | Tool | sports-data-collector |
8      | Video id | annotations (human_id) |
9      | Generated | 2026-06-09T12:35:01Z |
10     | Schema | 1.0 |
11
12     ## Video
13
14     | Field | Value |
15     | --- | --- |
16     | Duration | 58.1s |
17     | FPS | 59.94 fps (declared) |
18
19     ## Processing stages
20
21     **parse** ? `ok` (0.0s)
22     - rallies_parsed: 6
23     - _source_format_: cvat
24     - _fps_: 59.94
25     - _sampling_step_: 30
26
27     **validation** ? `ok` (0.0s)
28     - issues_blocker: 0
29     - issues_high: 0
30     - issues_medium: 0
31     - issues_low: 0
32     - n_blockers: 0
33     - _gate_: PASS
34
35     ## Findings & issues
36
37     _No findings._
38
39     ## Verdict
40
41     **PASS**
42
```

44. user (2026-06-12T09:06:30.333Z)

```
1      # Sports Data Collector ? Delivery / Ingestion Report (summary)
2
3      ## Run
4
5      | Field | Value |
6      | --- | --- |
7      | Video | annotations |
8      | Report generated | 2026-06-09T12:35:01Z |
9
10     ## Video
11
12     | Field | Value |
13     | --- | --- |
14     | Duration | 58.1s |
15     | Frame rate | 59.94 fps |
16
17     ## Rally summary
18
```

```
19 - **6** rallies
20 - **25.5s** of play
21 - **44%** of the video is rallies
22
23 ## Things to review
24
25 _Nothing needs your attention._
26
27 ## Verdict
28
29 This delivery passed all checks and is ready to import.
30
```

45. assistant (2026-06-12T09:06:32.635Z)

Let me check if there are more detailed reports or comparative eval data:

46. user (2026-06-12T09:06:33.168Z)

```
Exit code 123
ls: cannot access 'C:\Users\avidu\Projects\Annotation': No such file or directory
ls: cannot access 'Setup/Collect/Golden': No such file or directory
ls: cannot access 'Labelled/KushagraYashAviFirstVideo.report.md': No such file or directory
ls: cannot access 'C:\Users\avidu\Projects\Annotation': No such file or directory
ls: cannot access 'Setup/Collect/Golden': No such file or directory
ls: cannot access 'Labelled/KushagraYashAviFirstVideo.summary.md': No such file or directory
ls: cannot access 'C:\Users\avidu\Projects\Annotation': No such file or directory
ls: cannot access 'Setup/Collect/Golden': No such file or directory
ls: cannot access 'Labelled/MahadevpuraSingles.report.md': No such file or directory
ls: cannot access 'C:\Users\avidu\Projects\Annotation': No such file or directory
ls: cannot access 'Setup/Collect/Golden': No such file or directory
ls: cannot access 'Labelled/MahadevpuraSingles.summary.md': No such file or directory
ls: cannot access 'C:\Users\avidu\Projects\Annotation': No such file or directory
ls: cannot access 'Setup/Collect/Trajectories/roora/annotations.report.md': No such file or
directory
ls: cannot access 'C:\Users\avidu\Projects\Annotation': No such file or directory
ls: cannot access 'Setup/Collect/Trajectories/roora/annotations.summary.md': No such file or
directory
```

47. assistant (2026-06-12T09:06:35.040Z)

Let me search for more specific evaluation data comparing Roora with golden standard:

48. user (2026-06-12T09:06:35.660Z)

```
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md: $ **eval primitives**:
`Interval=Tuple[float,float]` (start<end). `Match{pred_index,gt_index,iou}`.
`MatchResult{matches, unmatched_pred_indices(=FP), unmatched_gt_indices(=FN)}`. `Score{iou_thres
hold,num_pred,num_gt,true_positives,false_positives,false_negatives,precision,recall,f1,mean_iou
,mean_start_error,mean_end_error}.as_dict()`.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\EVALUATION.md: Stage          Pred    TP
FP  FN  Prec  Rec    F1  mIoU  dStart  dEnd
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\EVALUATION.md: - **dStart / dEnd** ?
mean absolute boundary error in seconds (are rallies
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\EVALUATION.md: | Recall looks low only
at high `--iou` | Rallies *are* found but boundaries are loose ? check `dStart`/`dEnd`. Tune
`stitching` padding or the segmentation thresholds rather than the detector. |
```

49. assistant (2026-06-12T09:06:38.954Z)

Now let me check if there's any direct comparison data or notes in the DECISIONS file:

50. user (2026-06-12T09:06:39.266Z)

```

1      # Architecture Decision Records
2
3      ## ADR-001: Precise shuttle tracking via self-hosted TrackNetV4/WASB on WSL2
4
5      **Date:** 2026-05-25
6      **Status:** Accepted
7
8      ### Context
9
10     The pipeline uses an AI multimodal provider (Gemini) for *semantic* rally segmentation.
11     We need **precise frame-level detection of the moving shuttlecock** (and, generically,
12     balls for other sports) to produce higher-quality candidate rallies and a fine-tuning
13     signal.
14
15     We evaluated whether a commercially available model/API could do this. Findings:
16
17     - **Multimodal LLM APIs (Gemini, Claude, GPT-class) are the wrong tool for precise
18       localization.** They reason well about *what* is happening (rally vs. dead time) but
19       cannot reliably track a tiny, fast (~up to 400 km/h), motion-blurred shuttle frame by
20       frame.
21     - The precision leaders are **purpose-built trajectory models**, which are open source
and
22       self-hosted, not big-vendor APIs:
23     - **TrackNetV4** (TensorFlow) ? badminton-specific, heatmap + motion-attention,
24       highest reported shuttle accuracy; ships `src/predict.py` producing per-frame (X, Y)
25       coordinate CSV. https://github.com/AR4152/TrackNetV4
26     - **WASB** (PyTorch) ? multi-sport strong baseline (badminton/tennis/volleyball/?),
27       fits our `SportProfile` abstraction; needs a custom-video inference wrapper.
28       https://github.com/nttcom/WASB-SBDT
29     - True turnkey commercial APIs (Roboflow hosted models) exist but cap below a tuned
30       TrackNet; enterprise systems (Hawk-Eye, SkillCorner, TRACAB) are not API-accessible.
31
32     ### Decision
33
34     1. Adopt **self-hosted TrackNetV4** as the primary precise shuttle detector, with
**WASB**
35       as the multi-sport fallback.
36     2. Run these models under **WSL2 (Ubuntu)**, **not** native Ubuntu. The entire indexer
37       already runs on Windows; WSL2 keeps a single dev environment and quarantines the
38       TensorFlow/PyTorch dependencies. Native Ubuntu would require dual-boot and migrating
39       the whole project for marginal gain.
40     3. Integrate via a **subprocess adapter**: a detector module that shells out to the WSL
41       `predict.py`, reads the coordinate CSV, and converts trajectory ? candidate rallies
42       feeding the existing stitching flow. This keeps the TF dependency fully isolated from
43       the main Python environment.
44
45     ### Consequences
46
47     - **WSL I/O caveat:** `/mnt/c` access is slow. Stage match videos inside the WSL
48       filesystem (e.g. `~/clips/`) before inference rather than reading large mp4s across
the
49       Windows?WSL boundary.
50     - TrackNetV4 (TensorFlow) and WASB (PyTorch) live in **separate conda envs**.
51     - Next step after integration: manual video annotation ? fine-tune the detector.
52
53     ---
54
55     ## ADR-002: Use the sports-data-collector's rally annotations as a golden eval +
temporal-segmenter training set
56
57     **Date:** 2026-05-26
58     **Status:** Accepted
59
60     ### Context
61

```

```

62 A sibling project, sports-data-collector, curates sports videos and stores per-rally
63 `[start, end)` timestamps (plus stroke counts, scores) in an SQLite DB, with a
64 commercial-license flag per source. We asked whether this data could improve the
65 indexer's rally detection, and for which task.
66
67 Key findings:
68
69 - The annotations are temporal-only (when rallies start/end). They do not
contain
70 per-frame shuttle `(x, y)` coordinates, so they cannot train the shuttle detector
71 (TrackNetV4/WASB ? see ADR-001), which needs coordinate labels.
72 - They are an excellent golden evaluation set and a training signal for the
73 temporal rally segmenter (active vs. dead play over time) ? the stage that is
today
74 a hand-tuned heuristic, not a learned model.
75 - The collector's five sports tables (`badminton_rallies`, `tennis_points`,
76 `cricket_deliveries`, `pickleball_rallies`, `tabletennis_rallies`) all share a
77 `(video_id, ordinal, start_time, end_time)` shape, and the indexer already has
78 `sport_registry` / `segmenter_registry`. So a generic `(start, end)` bridge serves
79 every sport.
80 - The two repos key videos differently: the collector by a human `video_id`, the indexer
81 by file MD5. A re-encode/re-download changes the bytes, hence the MD5.
82
83 ### Decision
84
85 1. Treat the collector's rally timestamps as (a) the golden eval set for rally
86 segmentation and (b) the labels for training a temporal segmenter ? explicitly
87 not as shuttle-detector training data.
88 2. Bridge the repos with a sport-agnostic export (`curate export` in the collector):
89 per-video `start,end` CSVs (the indexer's eval format) + a `manifest.json` that pairs
90 each CSV with its video file. Gate on `is_commercial` so only commercially usable
data
91 flows through.
92 3. Honour the MD5 keying contract: evaluation/processing must run on the exact video
93 file recorded in the manifest; acquisition (ADR-003) therefore precedes processing.
94
95 ### Consequences
96
97 - The collector becomes the producer of labels; the indexer the consumer (processing
+
98 scoring + training). Neither takes a hard dependency on the other beyond the manifest.
99 - Scoring is free and deterministic (reads the DB; no API). See `docs/EVALUATION.md`.
100 - Only invest in frame-level coordinate labels (for the detector) if evaluation
shows
101 detection, not segmentation, is the bottleneck.
102
103 ---
104
105 ## ADR-003: Video acquisition ? always best resolution, upgrade in place, authenticated
cookies
106
107 **Date:** 2026-05-26
108 **Status:** Accepted
109
110 ### Context
111
112 The collector's initial crawl downloaded videos with `yt-dlp -f worstvideo` (256x144),
113 mislabelled as 1080p. The indexer needs 720p MP4 for meaningful YOLO/shuttle detection.
114 Re-acquiring full-resolution sources surfaced several hard-won realities:
115
116 - The ShuttleSet corpus tops out at 1080p (no 4K); "best available" resolves to
1080p.
117 - Authenticating with a YouTube Premium account is required for reliable,
unthrottled
118 downloads ? anonymous downloads are throttled (~1.3 MiB/s) and non-deterministic

```

```

119     (identical commands variously yielded 1080p DASH, 1080p HLS, or a 144p fallback).
120 - **Chrome** cookies can't be read on Windows (app-bound encryption / DPAPI, yt-dlp
121   #10927); **Firefox** cookies read cleanly even while open.
122 - The `web`/`web_safari` player clients are forced into **SABR streaming**, which strips
123   download URLs and silently drops to 144p (yt-dlp #12482). yt-dlp's `default` client
124   serves real downloadable 1080p DASH with auth cookies.
125 - Bulk bursts trip YouTube **rate-limiting** (a 22-video burst failed all at once,
though
126   each worked in isolation).
127
128   ### Decision
129
130   1. **Always fetch the best resolution available** (`bv*+ba/b -S res,...`, merged to
MP4);
131     ffprobe the result and store the *actual* width/height/fps ? never assume.
132   2. **Upgrade in place, never clean-slate:** the `fetch` command re-downloads a video and
133     updates only `local_path`/`resolution`/`fps`, preserving every rally annotation and
134     curation status (deleting would cascade-delete the valuable labels).
135   3. Use **authenticated cookies** (`--cookies-from-browser firefox` / `--cookies file`)
and
136     the **`default` player client**; enable the EJS challenge solver (deno).
137   4. **Throttle bulk runs** (`--sleep` + per-video `--retries`/backoff`) and treat a
138     below-min-height download as a **failure** (discard, don't commit) so a flaky low-res
139     result never overwrites a good entry.
140
141   ### Consequences
142
143   - All 22 fetchable ShuttleSet matches are now true 1080p (one match, shuttleset_12, has
no
144     source URL in the catalog and remains video-less but keeps its annotations).
145   - Acquisition is reproducible and resumable; the ~26 GB corpus is kept outside git.
146   - Future non-YouTube or 4K sources are supported by the same code (optional
`max_height`).
147
148   ---
149
150   ## ADR-004: Add an offline, API-free *learned* rally segmenter as a first-class mode
151
152   **Date:** 2026-05-26
153   **Status:** Proposed (substrate pending data)
154
155   ### Context
156
157   The default high-quality path (`yolo_hybrid`) refines YOLO candidate windows with
**Gemini**,
158   which costs money and requires network access. With a now-1080p, commercially-licensed,
159   rally-annotated corpus (ADR-002/003), we can learn the refinement step offline instead.
160
161   Crucially, the **API-free signal already exists and is already persisted**:
`yolo_hybrid`
162   Stage 1 is pure YOLO (player tracking ? per-window motion features: peak/mean velocity,
163   active-frame count, track-id count) stored in `candidate_segments.stats`. Gemini is only
164   the Stage-2 refiner. So the rally annotations can **train a classifier to replace
Gemini**,
165   with no external API and **no new labeling**.
166
167   Three candidate substrates were weighed:
168
169   - **Tune `motion` thresholds** on the annotations ? trivial, fully offline, but a low
170     ceiling (pixel motion can't separate rally from crowd/pan/replay). Use as a *floor*.
171   - **YOLO features ? trained classifier** ? reuses already-stored, semantically
meaningful
172     player-motion features; medium-high ceiling; moderate effort. *Recommended primary.*
173   - **WASB shuttle-motion ? rally detector** ? highest ceiling (the shuttle defines play),
174     fully offline but heaviest (needs the WSL2 WASB runner, ADR-001); can't train WASB

```



```

175         itself (no xy labels), only a rally layer on its output.
176
177     ### Decision
178
179     1. Build an offline, API-free learned segmenter as a new registry mode (e.g.
180         `yolo_learned`): label detector windows by overlap with GT rallies, train a
lightweight
181         classifier on the persisted features, emit final segments with zero API calls,
and
182         score it with the existing harness against `motion` and a Gemini reference.
183     2. Choose the substrate from data, not assumption. A learned refiner can only
sharpen
184         the windows the detector found, so the deciding metric is the detector's candidate-
stage
185         recall vs. the annotations: high recall ? build on YOLO features; low recall ?
bring in
186         WASB's shuttle signal to raise the ceiling.
187     3. Keep the Gemini-backed `yolo_hybrid` as an upper-bound reference, not a
production
188         dependency.
189
190     ### Consequences
191
192     - The final tool gains a self-contained, offline, free rally mode trained on its own
193         curated data ? runnable air-gapped, improvable as more matches are annotated.
194     - The approach is sport-generic: the same label?feature?classifier machinery extends
to
195         tennis/cricket once those have annotations.
196     - The full paid 22-video Gemini evaluation is not run; spend is limited to a single
197         reference video.
198
199     ### Update (2026-05-26): Phase 3 evidence ? substrate is WASB, not YOLO
200

```

51. assistant (2026-06-12T09:06:41.404Z)

Let me check the rest of the DECISIONS file and also look more specifically at the Pickleball review to extract the exact timing offset data:

52. user (2026-06-12T09:06:42.519Z)

Thank you for completing the pilot trio ? both deliveries ingest cleanly, shot counts are filled and plausible, and numbering and rally separation are clean. However, this round has **one serious issue: rally boundary accuracy.** Every rally's start/end in both videos is off by **0.5 s or more ? up to ~2 s in several** ? against the agreed tolerance of **±0.3 s** (Brief §4, Guide Part 3). The badminton round's timings were generally good, so this is a regression; we'd like to understand what changed in the process. The sections below explain the impact, the root cause we found in the files, and the conventions to lock before the Phase-B batches.

2. Boundary accuracy ? outside the agreed tolerance ***(most important)***

What we found. A full frame-by-frame re-check of all 12 rallies shows every `start`/`end` boundary off by 0.5 s or more, several by up to ~2 s.

Root cause (from your files). All 24 rally boundaries sit **exactly** on the 30-frame sampling grid ? at 59.94 fps that grid is ~0.5 s apart, so a boundary placed only on grid frames **cannot** meet ±0.3 s. Sampling every 30th frame **inside** a rally is fine; but the **first box** must be on the exact serve-contact frame and the last box on the exact ball-dead frame (Guide Part 3), even when those fall between grid steps.

Why this matters so much. These labels are the golden ground truth we train and evaluate a rally-segmentation model against ? **boundary precision is the product itself**. The rallies in these videos run 1?8.5 s, so a 0.5?2 s boundary error corrupts 10?100% of a rally's span; the 1.0 s table-tennis rally #2 can be erased entirely by it. Concretely: at our standard evaluation

gate (overlap ? 50%), a typical 4.5 s rally labeled 2 s late ****fails to match even a perfectly correct detection**** ? the dataset would score correct model output as wrong and wrong output as right. Imprecise gold is worse than no data, because it silently corrupts every model trained on it and every quality number reported from it.

****Going forward:**** place the first/last box of every rally on the exact contact/dead frame (frame-step to find it ? Guide Parts 3 and 6), and self-check boundaries against ± 0.3 s before submitting (Guide Part 7 checklist).

3. Rallies already in progress when the video starts ? skip them

Pickleball ****rally #1 starts at the very first frame of the file (0:00.000)****. The serve contact is not in the video ? the file begins mid-rally ? so its start time is unknowable and the rally should not have been labeled at all.

****Convention to follow from now on:**** if a video begins during an ongoing rally, ****skip that rally entirely and start annotating from the first rally that starts (serve contact visible) within the video.**** Never extrapolate or guess a boundary you did not see. The same applies at the end of a file: if the video cuts off mid-rally, skip that rally. When in doubt, flag it to us rather than guessing (Guide Part 6).

4. Reminder for Phase B ? dead time and warm-up

Phase-B videos are ~30 minutes and will contain substantial non-match footage, so we want to re-emphasize two rules now (Guide Rule 1 + Part 6):

- ****Skip all dead time**** ? before the match, between rallies, during breaks: walking, picking up the ball, serve preparation, changeovers, coaching. None of it is ever labeled.
- ****Skip warm-up / knock-up rallies and practice points**** ? they are not real rallies. Labeling starts at the ****first real, scored serve****. When in doubt whether the match has started, flag it to us.

5. Schema hygiene ? remove the fields we did not ask for

Please remove the following from the CVAT label schema; they add noise and, in one case, actively confusing data:

53. assistant (2026-06-12T09:06:45.174Z)

Perfect! Now let me check if there are any more details or analysis about the cause. Let me create a comprehensive summary now:

54. user (2026-06-12T09:07:03.650Z)

Structured output provided successfully

55. assistant (2026-06-12T09:07:11.789Z)

Perfect! I have completed a thorough search across all three repos and found the comprehensive documentation of the Roora annotation delay issue. Let me provide a clear summary:

Summary of Findings

I found ****extensive documentation**** of the Roora annotation vendor's timing delay across the three repos. The key findings are:

Where It's Documented

1. ****Primary source:**** `C:\Users\avidu\Projects\Annotation Setup\Roora Pilot Review ? Pickleball & Table Tennis (2026-06-10).md`
 - Formal review document dated June 10, 2026 following the pilot delivery

2. ****Secondary source:**** `C:\Users\avidu\Projects\Annotation Setup\Roora Pilot Review ? Badminton (2026-06-06).md`
 - Earlier review showing badminton had acceptable timing (establishing this was a regression)
3. ****Specification references:****
 - `C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ROORA\_BRIEF.md` (vendor spec)
 - `C:\Users\avidu\Projects\sports-data-collector\docs\annotation\ANNOTATION\_GUIDE.md` (detailed requirements)

Exact Measured Numbers

- ****Pickleball & Table Tennis:**** Every rally boundary (start and end) was off by ****0.5 seconds or more****, with ****several up to ~2 seconds****
- ****Direction:**** Both start AND end boundaries were delayed (not unidirectional)
- ****Tolerance requirement:**** ± 0.3 seconds (per Brief §4 and Guide Part 3)
- ****Impact:**** For 178.5 second rallies, this 0.5s error corrupts 100% of the rally span

Documented Root Cause

The analysis identified a specific technical cause: ****All 24 rally boundaries were placed exactly on a 30-frame sampling grid**** in CVAT. At the declared 59.94 fps, this grid = ~0.5 seconds apart. The annotators were sampling at fixed intervals rather than frame-stepping to exact serve-contact and ball-dead frames, causing boundaries to be snapped to the nearest grid point.

Documented Mitigation

Going forward, the guide requires:

1. Frame-step to find exact serve-contact (first frame) and ball-dead (last frame)
2. Self-check boundaries against ± 0.3 s tolerance before submission
3. Updated CVAT schema and process documentation

The badminton pilot had "generally good" timings, so this systematic delay was specific to the later pickleball and table tennis deliveries.